



有机化学

陈旭 副研究员 硕士研究生导师

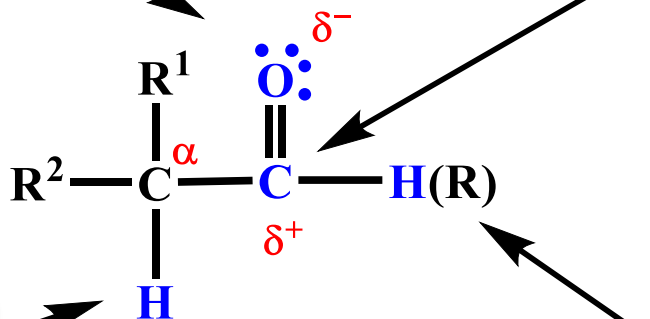
生工楼A429, xuchen@Jiangnan.edu.cn



醛、酮的化学性质

羰基氧有弱碱性
与酸结合

羰基碳有亲电性
亲核加成
氢化还原



α -H有酸性
烯醇化
 α -卤代
羟醛缩合反应

醛的氧化



教学安排

第1章	绪论
第2章	烷烃和环烷烃
第3章	不饱和烃和共轭效应
第4章	芳烃
第5章	立体化学
第6章	卤代烃
第7章	醇酚醚
第8章	醛和酮
第9章	羧酸及其衍生物
第10章	有机含氮化合物
第11章	生物大分子及其单体的结构与性质
第12章	有机化合物的波谱分析



第九章 羧酸及其衍生物

9.1 羧酸的分类和命名

9.2 羧酸的结构

9.3 羧酸的物理性质

9.4 羧酸的化学性质

9.5 羧酸的制法

9.5 羟基酸

9.6 羧酸衍生物的分类和命名

9.7 羧酸衍生物的物理性质

9.8 羧酸衍生物的化学性质

羧酸



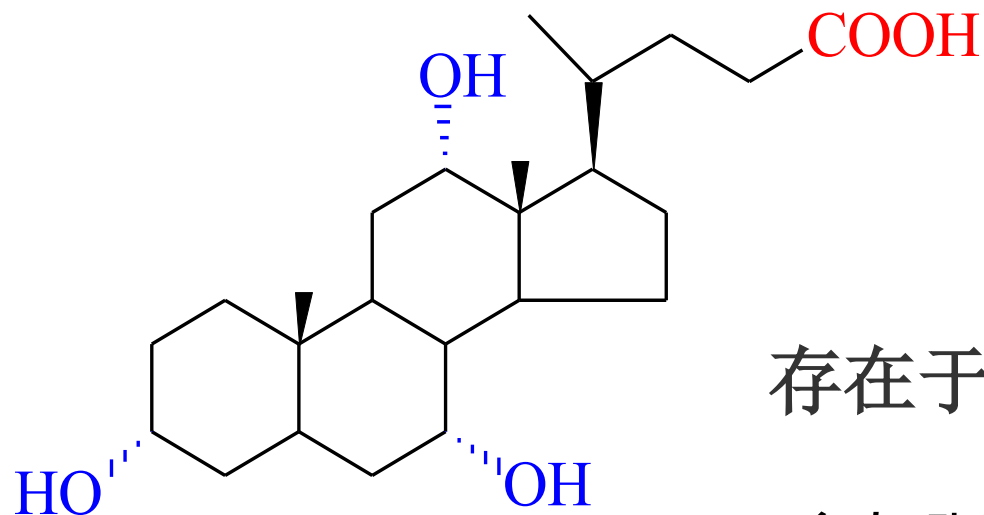
醋酸

Acetic acid

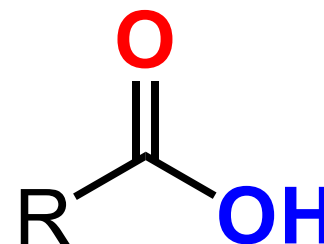


丁酸

Butanoic acid



胆酸 (Cholic acid)



存在于牛、羊、猪的胆汁中

参与脂质消化与吸收
调控胆固醇稳态.....

第九章 羧酸及其衍生物



9.1 羧酸的分类和命名

➤ 俗名（了解） 根据来源：



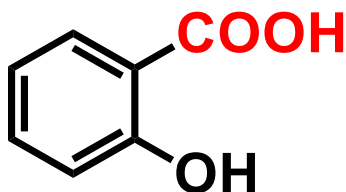
蚁酸
formic acid



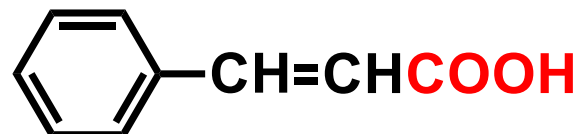
醋酸
acetic acid



草酸
oxalic acid



水杨酸
salicylic acid



肉桂酸
cinnamic acid

合成?

第九章 羧酸及其衍生物

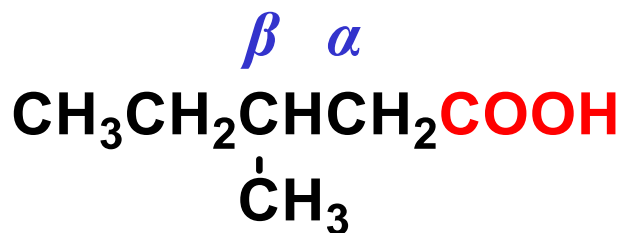


➤ 系统命名 (IUPAC命名) : 与醛类似

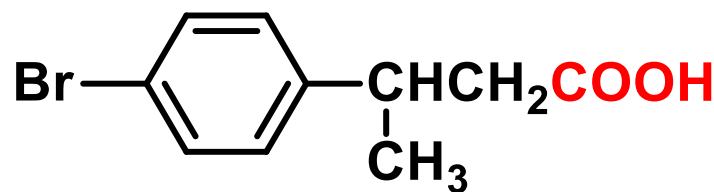
- ① 选择含**羧基**的最长碳链作为主链, 称“某酸”;
- ② 从**羧基碳**开始编号, 取代基位次也可用希腊字母 α , β , γ , δ 等表示, 最末端碳原子可用 ω 表示。



丙酸
propionic acid



3-甲基戊酸
(β -甲基戊酸)



3-(4-溴苯基)丁酸
3-(4-bromophenyl)butanoic acid

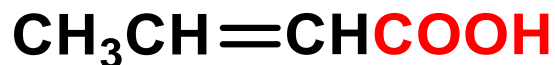
第九章 羧酸及其衍生物



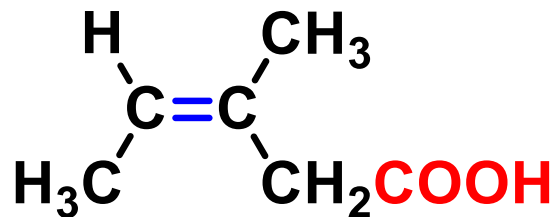
9.1 羧酸的分类和命名

✓ 不饱和脂肪酸:

选择同时含有**羧基**和**不饱和碳原子**在内最长碳链为主链，称为“某烯酸”或“某炔酸”，编号时从**羧基碳**开始。



丁-2-烯酸



(Z)-3-甲基-戊-3-烯酸

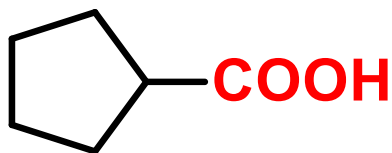


第九章 羧酸及其衍生物

9.1 羧酸的分类和命名

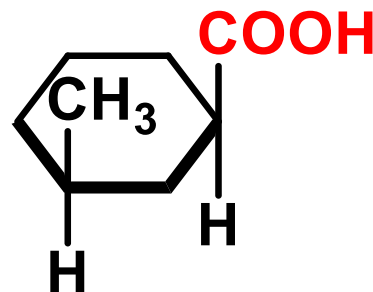
✓ 一元的脂环族羧酸：

可以将其看做甲酸衍生物，将环做为取代基。



环戊基甲酸

cyclopentanecarboxylic acid



(1*R*, 3*S*)-3-甲基环己基甲酸

(1*R*, 3*S*)-4-methylcyclohexane-
carboxylic acid

第九章 羧酸及其衍生物



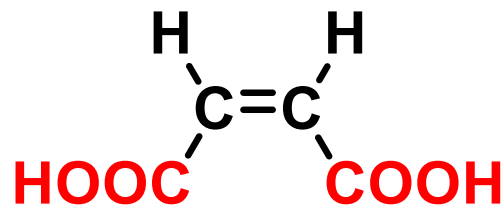
9.1 羧酸的分类和命名

✓ 多元羧酸:

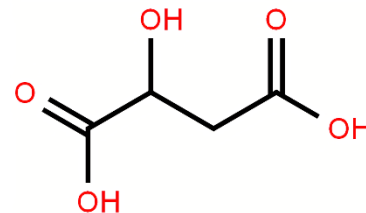
选择含有**尽可能最多羧基在内的最长碳链**为主链，并在“酸”前面标明羧基数目。



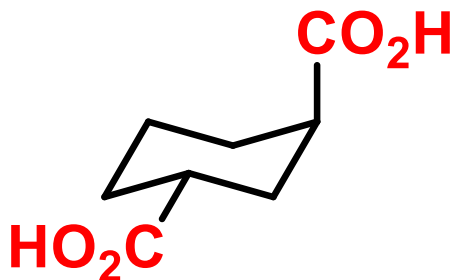
乙二酸



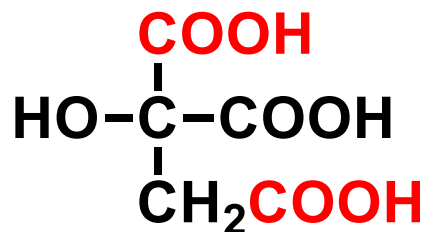
顺丁烯二酸
(马来酸, 缩苹果酸)



苹果酸



(1*R*,3*R*)-1,3-环己烷二羧酸



2-羟基-2-羧基丁二酸

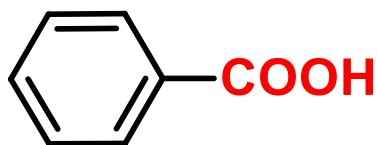
第九章 羧酸及其衍生物



9.1 羧酸的分类和命名

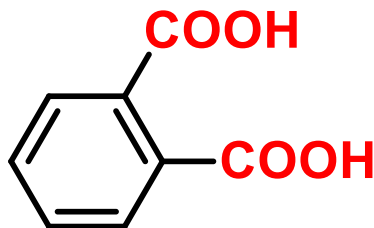
✓ 芳香酸：

以苯甲酸为母体，加上其它取代基的位次和名称，编号从羧基所连碳开始。



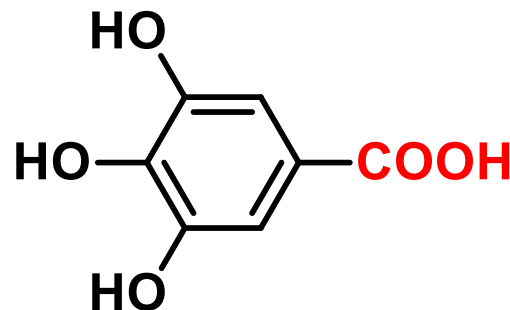
苯甲酸

benzoic acid



邻苯二甲酸

o-benzenedicarboxylic acid



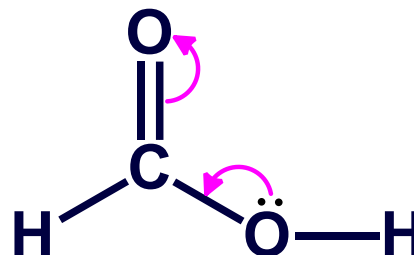
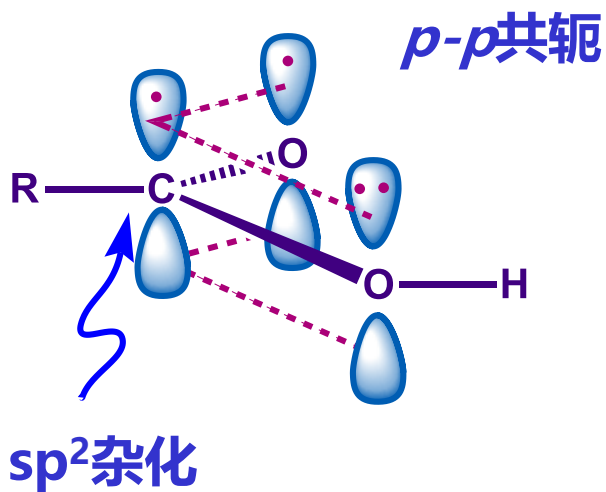
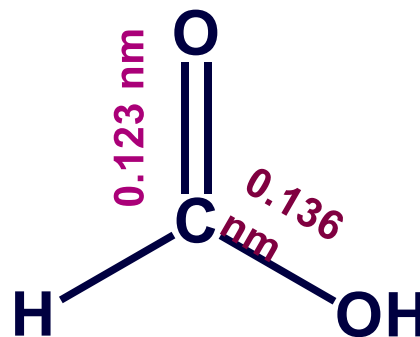
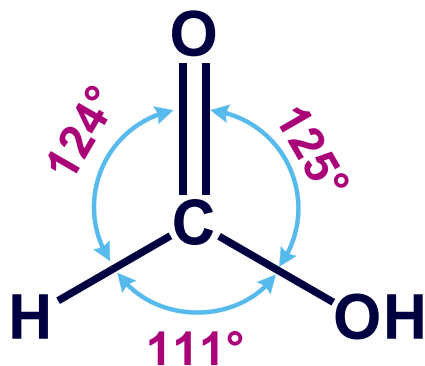
3,4,5-三羟基苯甲酸
(没食子酸)

3,4,5-trihydroxybenzoic acid

第九章 羧酸及其衍生物



9.2 羧酸的结构



第九章 羧酸及其衍生物

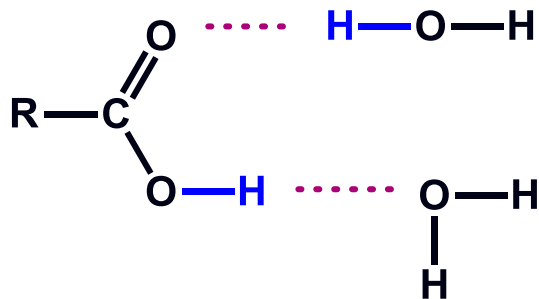


9.3 羧酸的物理性质

液体：甲酸~壬酸；

固体：癸酸以上脂肪酸、二元羧酸和芳香族羧酸

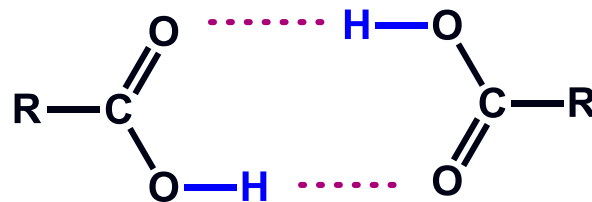
羧酸的**沸点**：比相对分子质量相同醇沸点高。



b.P(°C)

HCOOH 100.7

C₂H₅OH 78.5



b.P(°C)

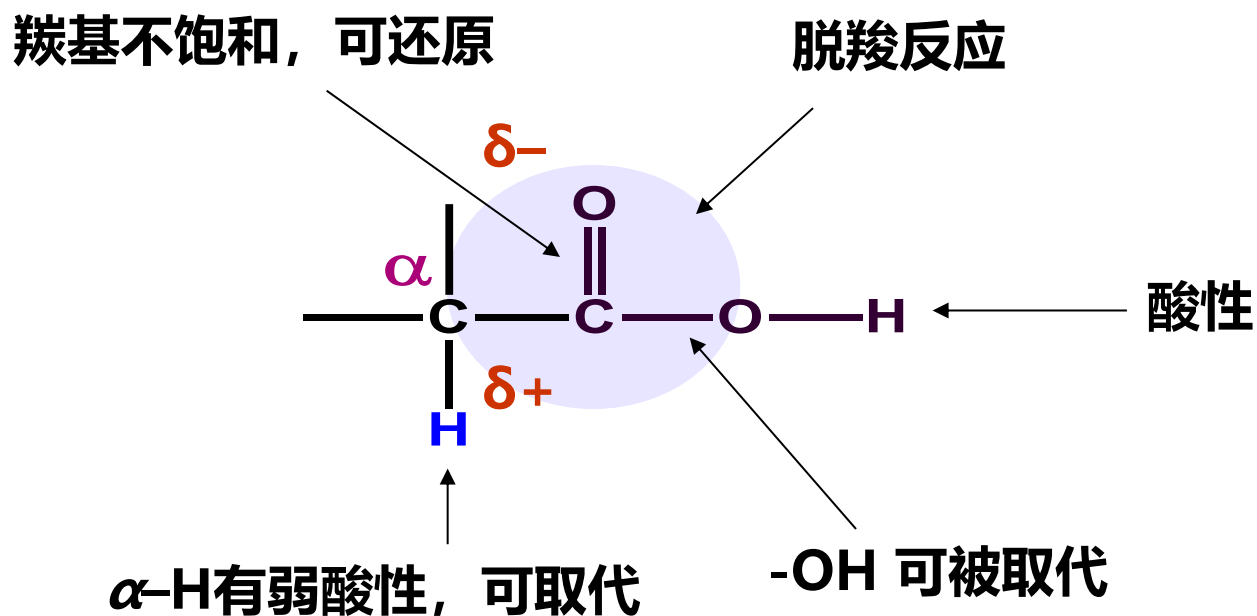
CH₃COOH 117.9

n-C₃H₇OH 97.4

第九章 羧酸及其衍生物



■ 羧酸的结构及化学性质分析





9.4 羧酸的化学性质

pK_a

9.4.1 羧酸的酸性



$R-\overset{\overset{O}{\parallel}}{C}-O-H$	4 ~ 5
$HO-\overset{\overset{O}{\parallel}}{C}-O-H$	6.38
 -O-H	10
$H-O-H$	15.74
$R-CH_2-O-H$	16 ~ 19
$R-C \equiv C-H$	~ 25
$R-\overset{\overset{H}{\mid}}{N}-H$	~ 34
$R-H$	~ 50

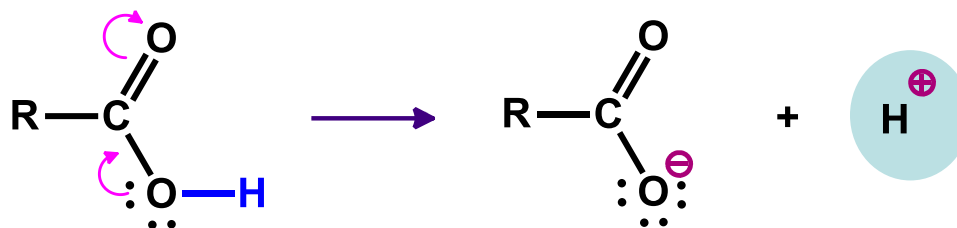




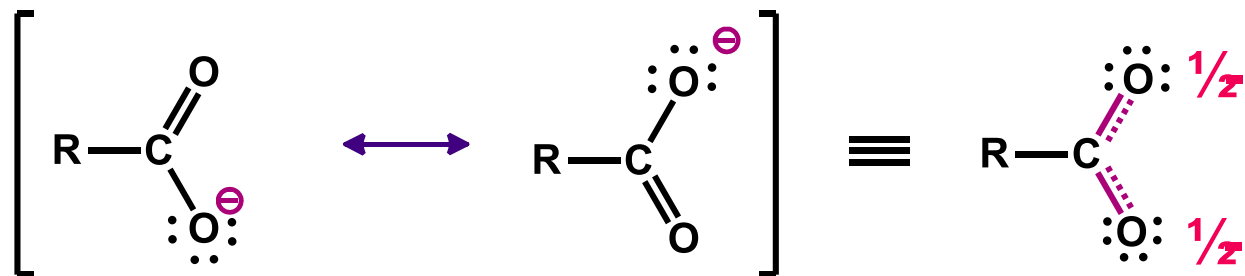
9.4 羧酸的化学性质

9.4.1 羧酸的酸性

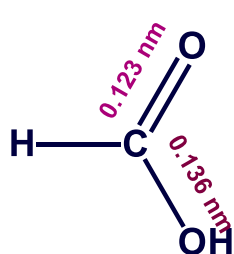
◆ 酸性



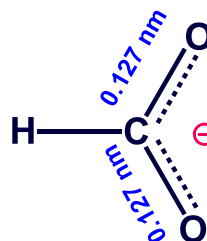
◆ 共振



◆ 结构



甲酸



甲酸根

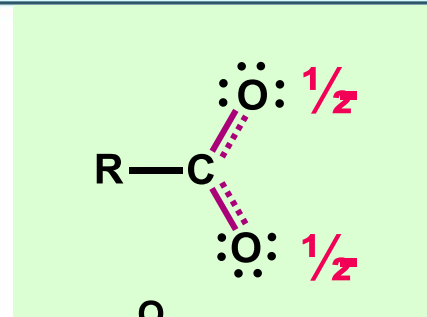
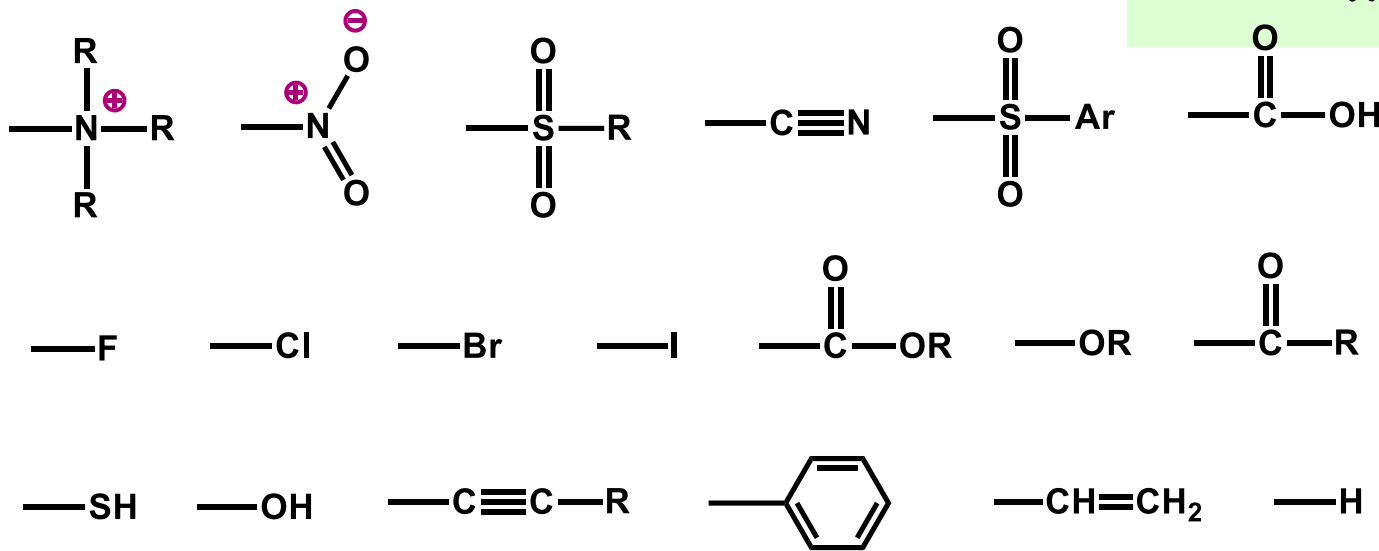


9.4 羧酸的化学性质

诱导效应

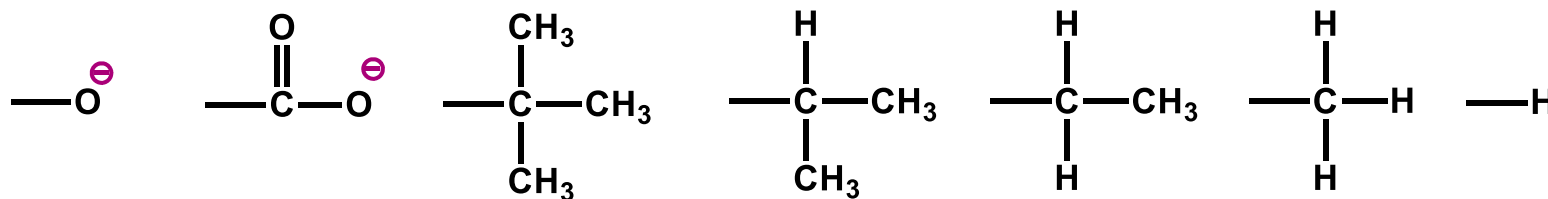
R

- / 吸电诱导效应



R

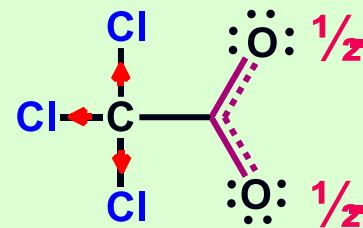
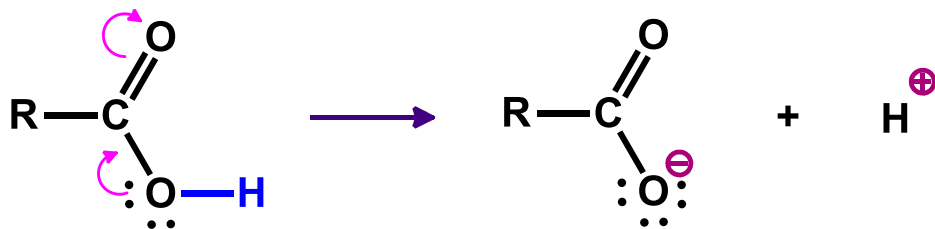
+ / 供电诱导效应





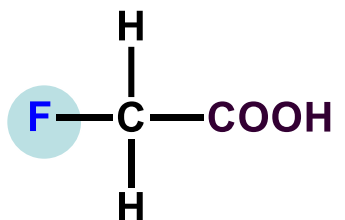
9.4 羧酸的化学性质

9.4.1 羧酸的酸性

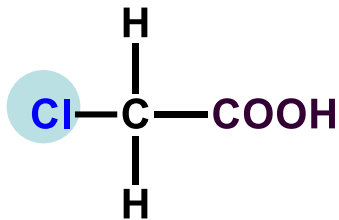


- 吸电诱导效应使负电荷得到分散，稳定了羧酸根负离子。
- 电负性愈强，负电荷分散程度愈大，负离子愈稳定。

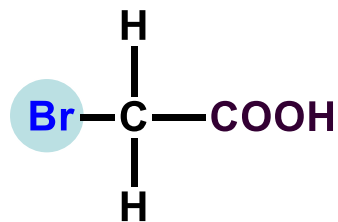
不同原子



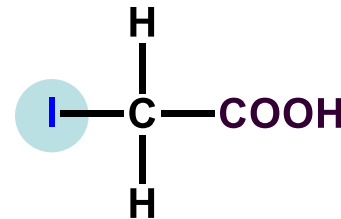
2.66



2.81



2.87

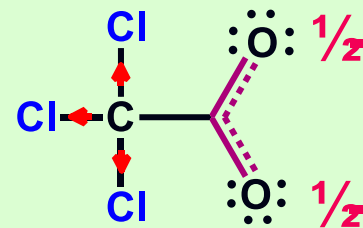
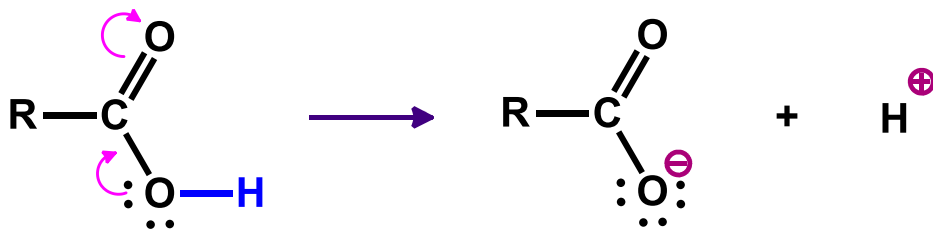


3.31



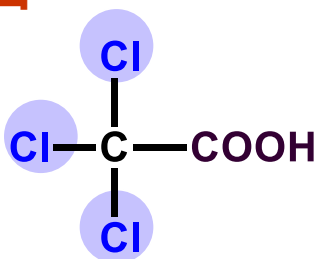
9.4 羧酸的化学性质

9.4.1 羧酸的酸性

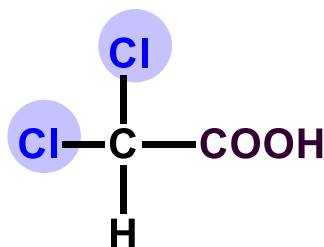


- 吸电诱导效应使负电荷得到分散，稳定了羧酸根负离子。
- 电负性愈强，负电荷分散程度愈大，负离子愈稳定。

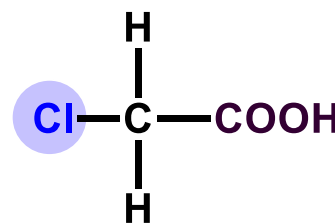
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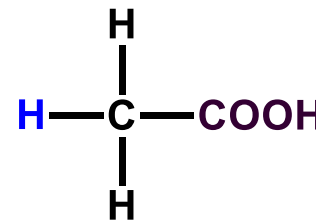
0.70



1.29



2.81

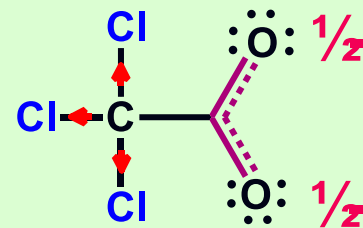
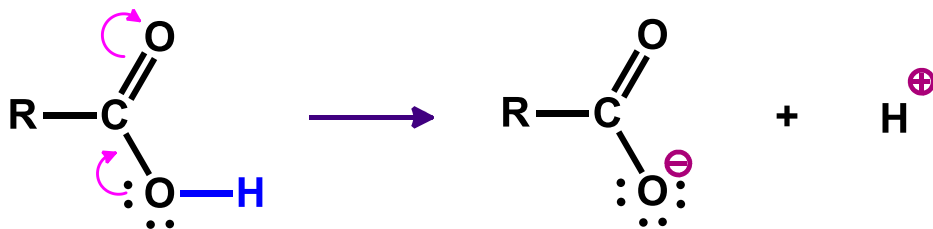


4.75



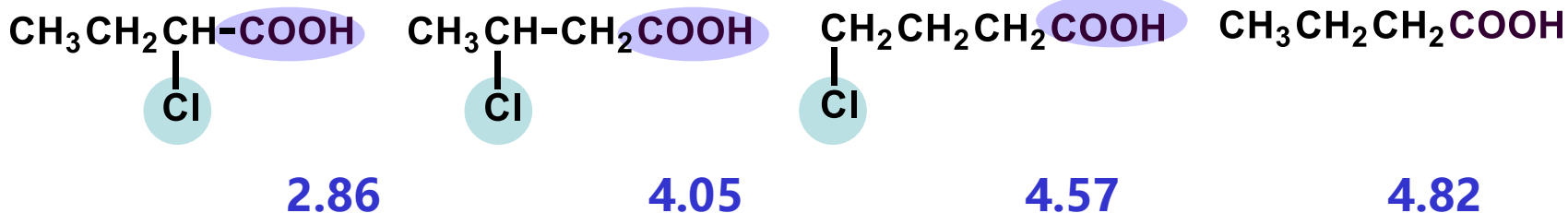
9.4 羧酸的化学性质

9.4.1 羧酸的酸性



- 吸电诱导效应使负电荷得到分散，稳定了羧酸根负离子。
- 电负性愈强，负电荷分散程度愈大，负离子愈稳定。

不同距离 诱导效应随着传递距离的增大而迅速减小。

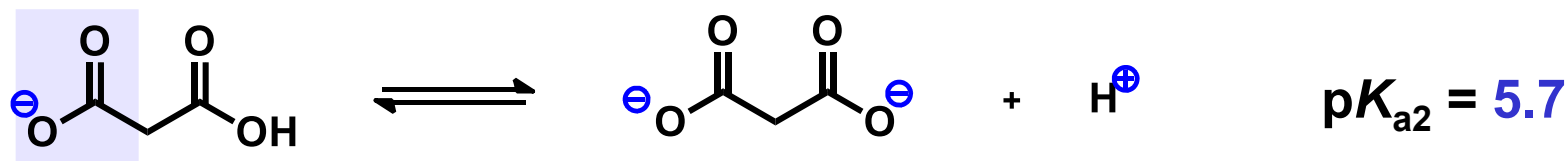
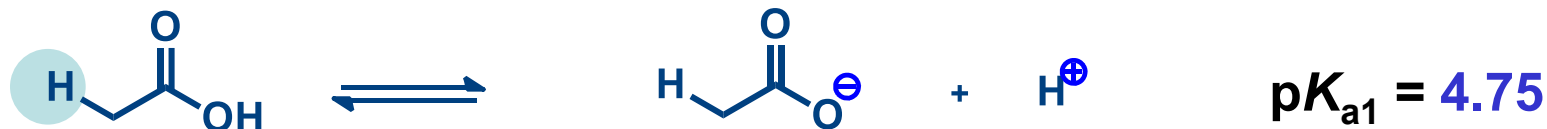
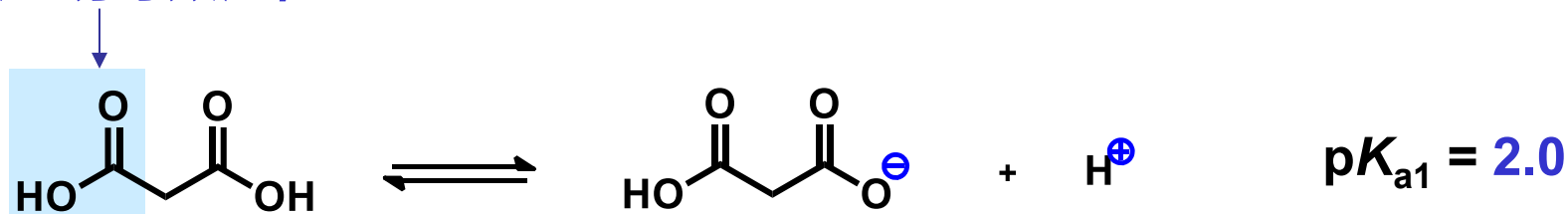




9.4 羧酸的化学性质

二元酸的酸性

吸电诱导效应, -I

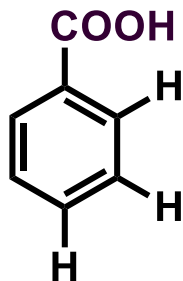


供电诱导效应, +I

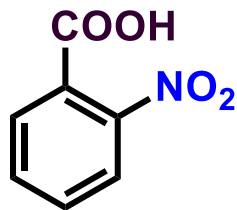
羧基的 -I 效应利于羧酸的第一次解离。



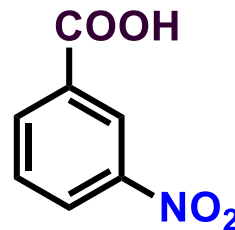
9.4 羧酸的化学性质



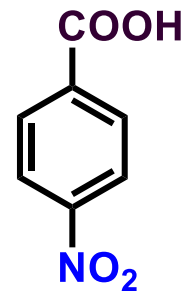
4.20



2.21



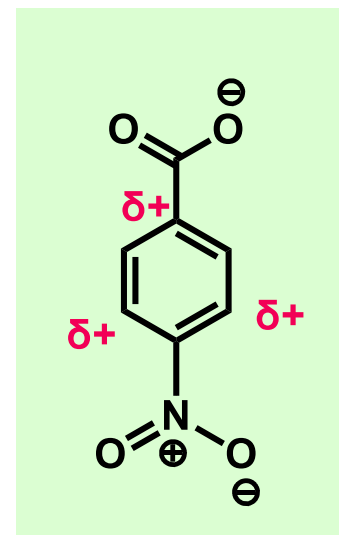
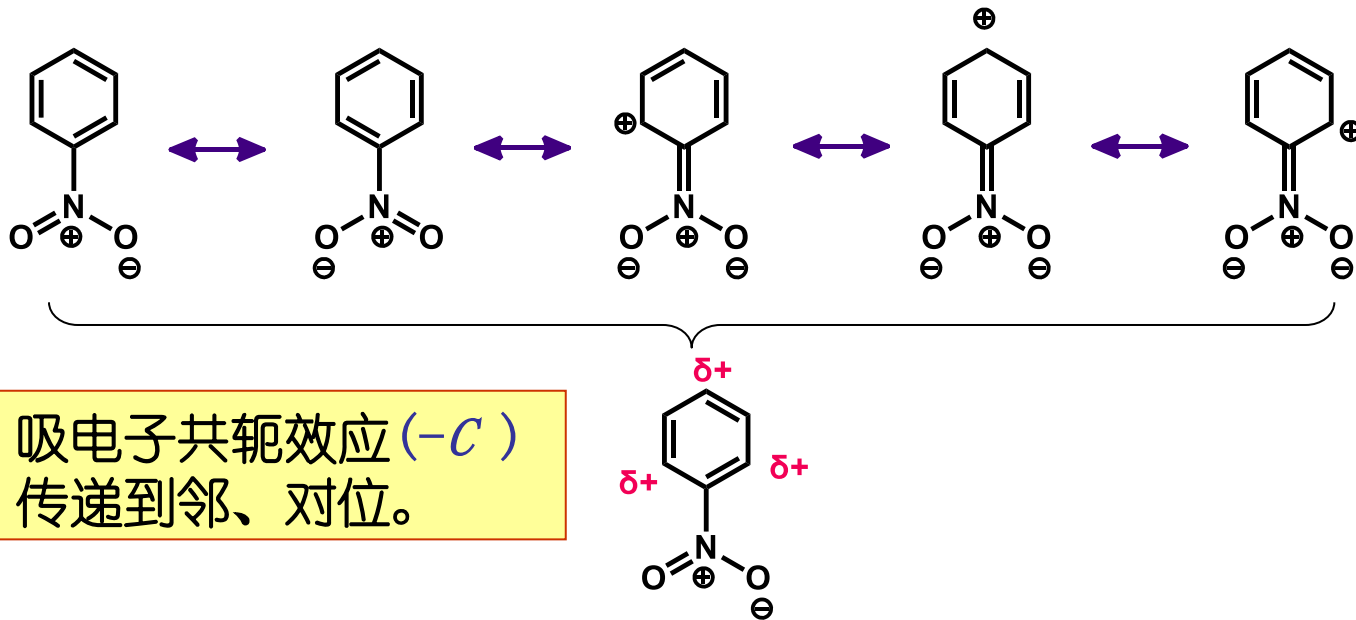
3.49



3.42

吸电子诱导效应(-I),又具有吸电子共轭效应(-C)

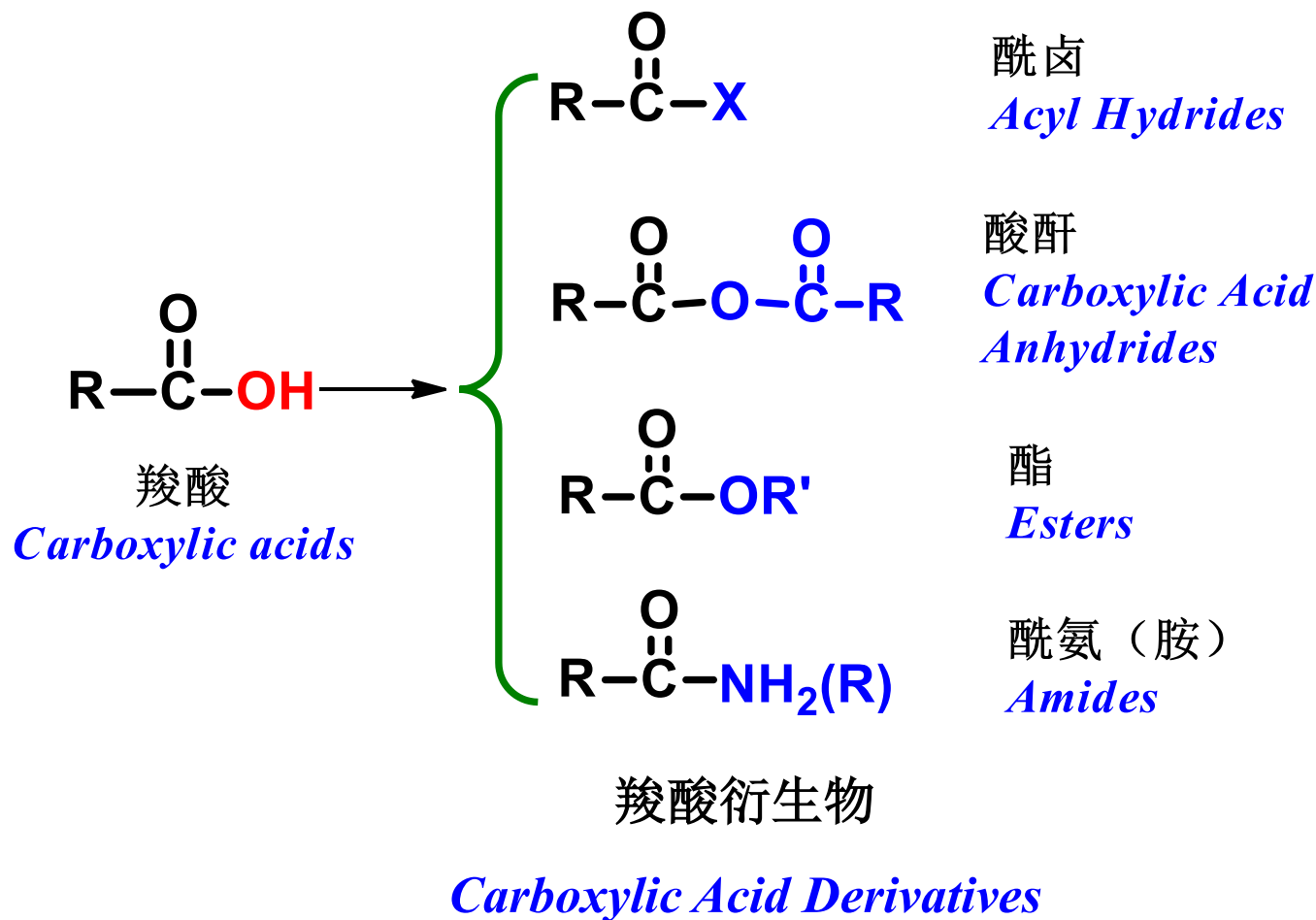
吸电子共轭效应(-C)





9.4 羧酸的化学性质

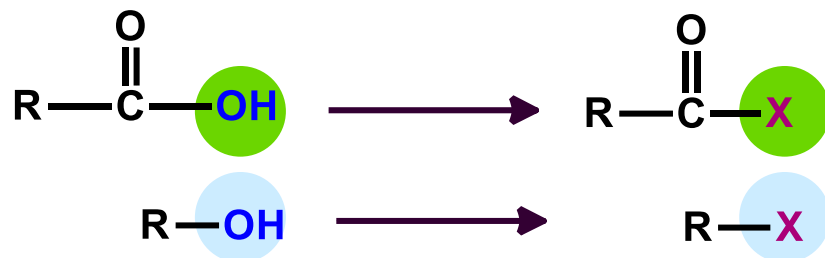
9.4.2 羧酸衍生物的生成（羟基被取代）



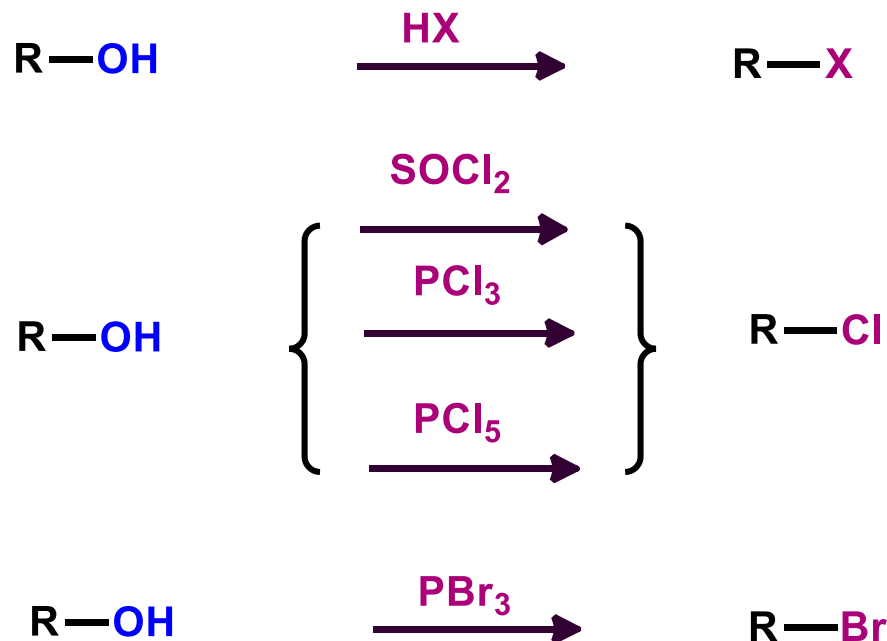


9.4 羧酸的化学性质

羧酸衍生物的生成-酰氯



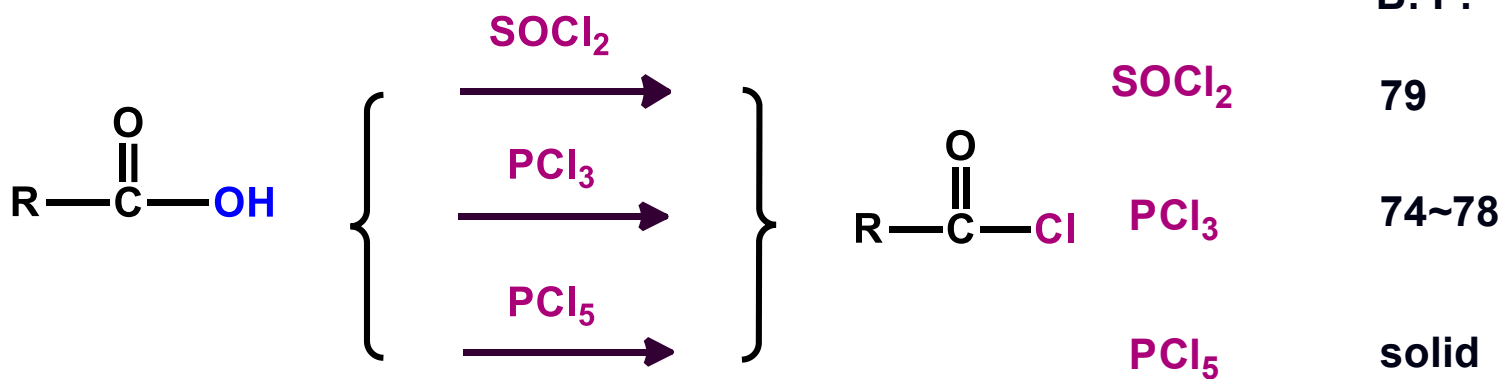
回顾: 将醇转变成卤代烃





9.4 羧酸的化学性质

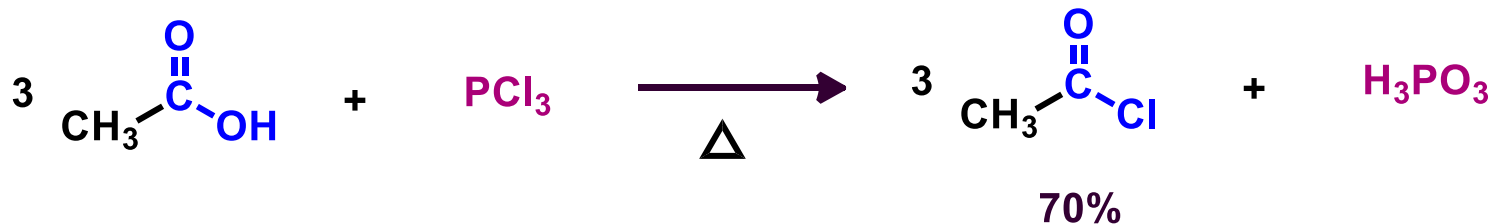
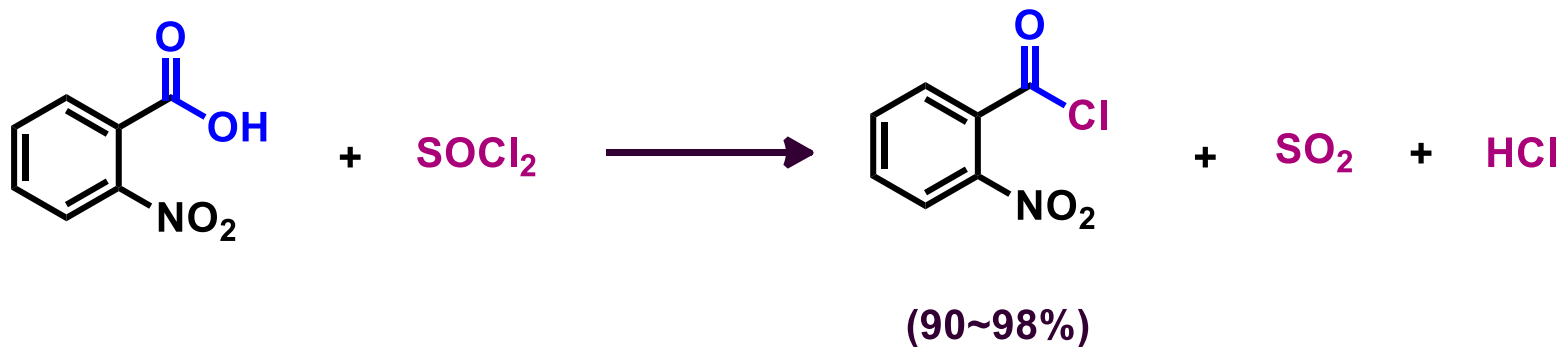
羧酸衍生物的生成-酰氯





9.4 羧酸的化学性质

羧酸衍生物的生成-酰氯

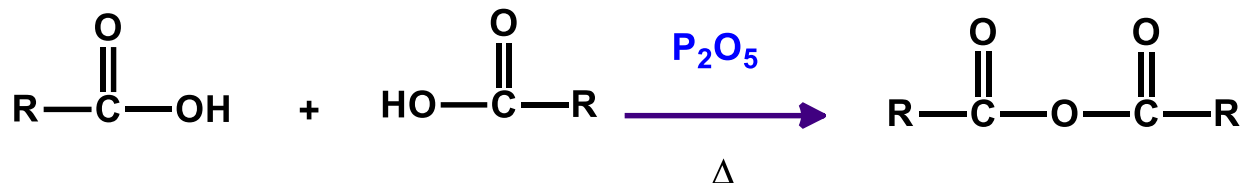




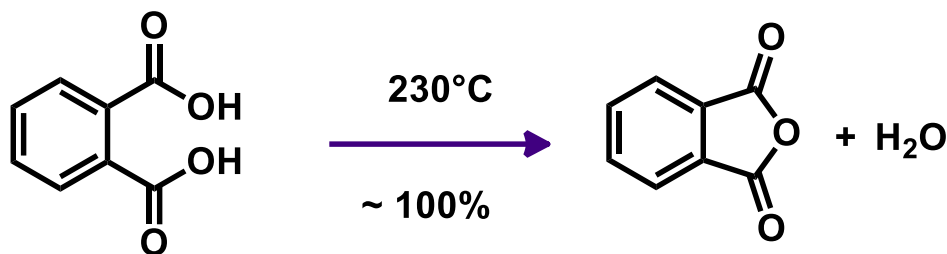
9.4 羧酸的化学性质

羧酸衍生物的生成-酸酐的生成

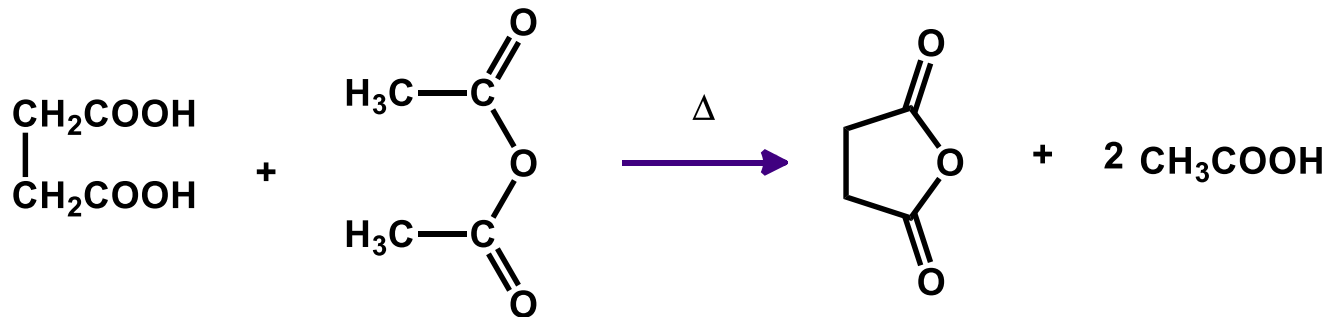
■ 脱水剂



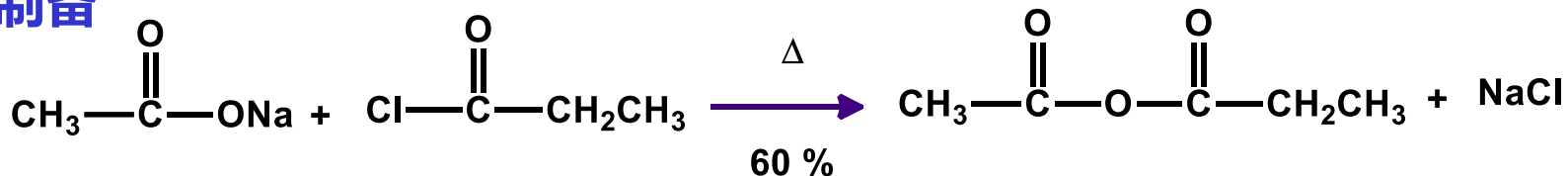
■ 加热



■ 交换



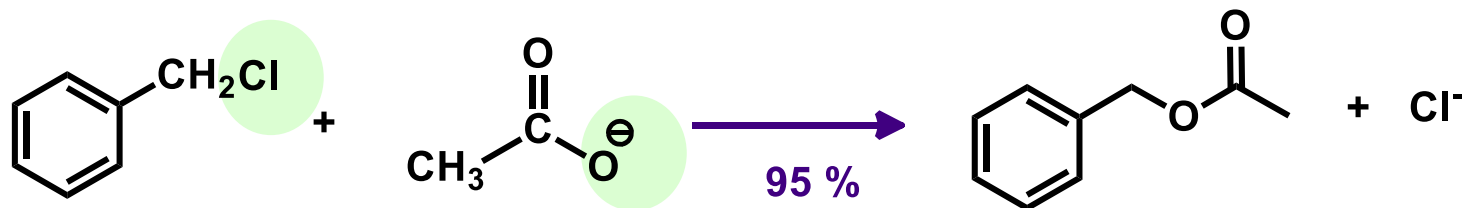
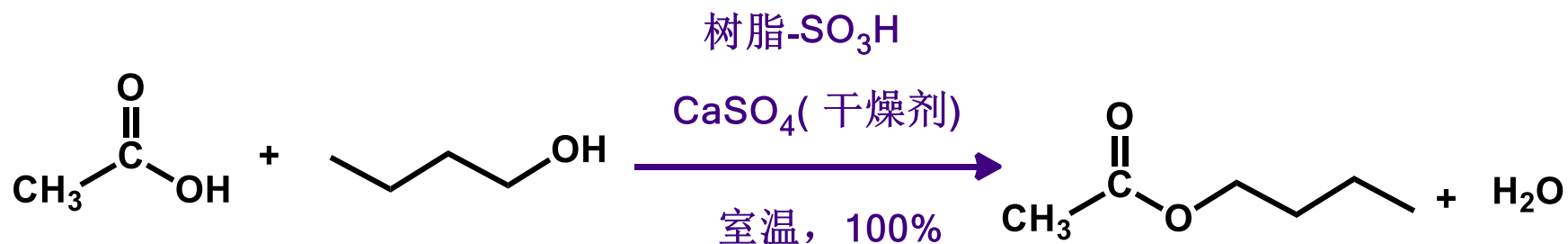
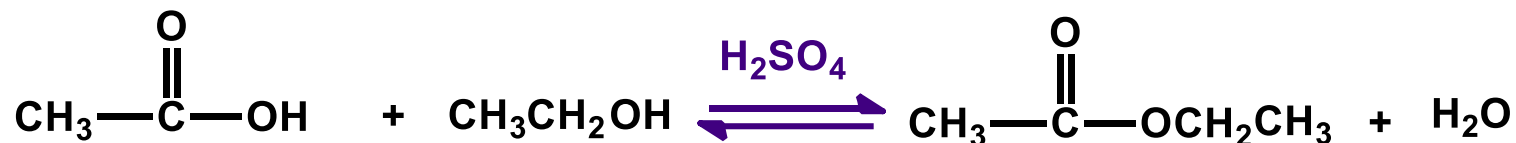
■ 混酐制备





9.4 羧酸的化学性质

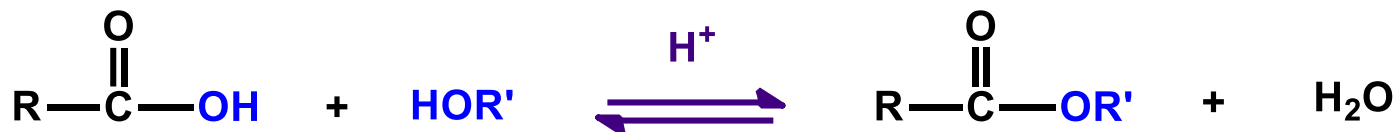
羧酸衍生物的生成-酯的生成





9.4 羧酸的化学性质

羧酸衍生物的生成-酯的生成

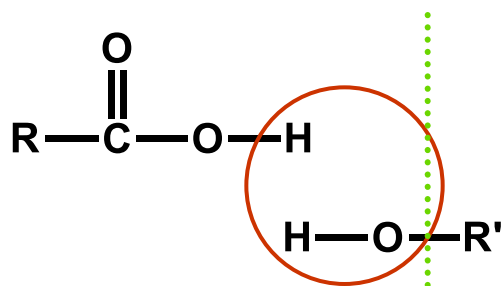


酯化反应特点：

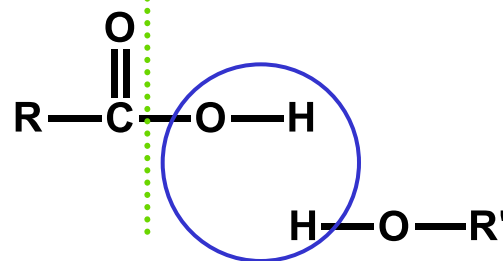
- 反应需要 H^+ 催化，无催化剂时反应很慢。
- 反应可逆（加大反应物用量或除去水使酯的产率提高）

两种可能

烷氧断裂



酰氧断裂

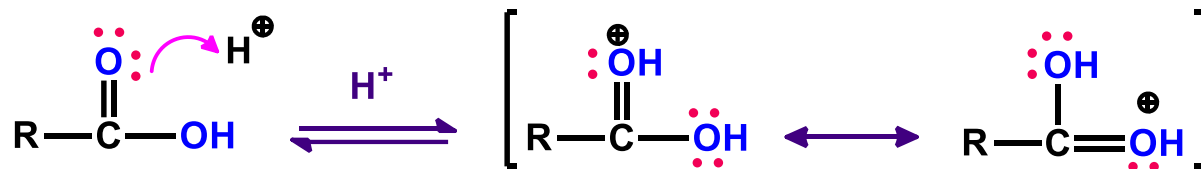




9.4 羧酸的化学性质

酰氧键断裂机理

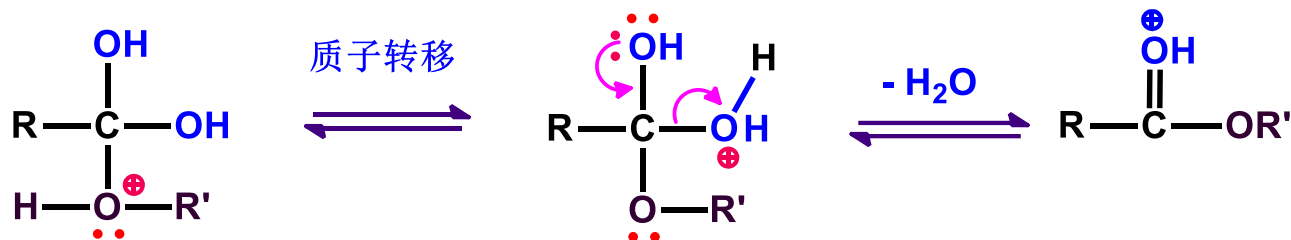
① 羰基质子化



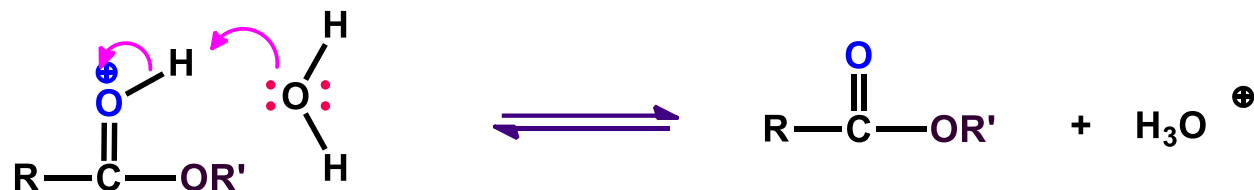
② 亲核进攻



③ 质子转移及脱水



④ 脱质子

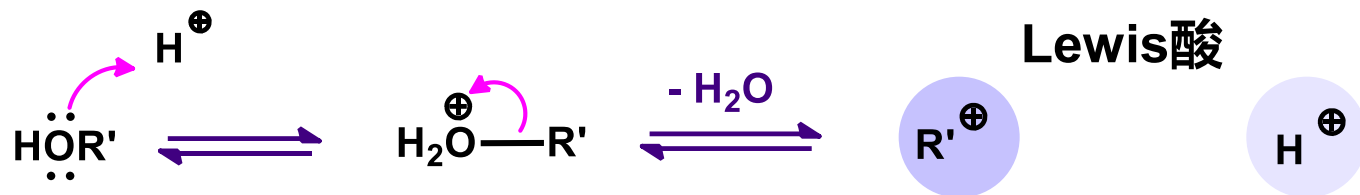


9.4 羧酸的化学性质



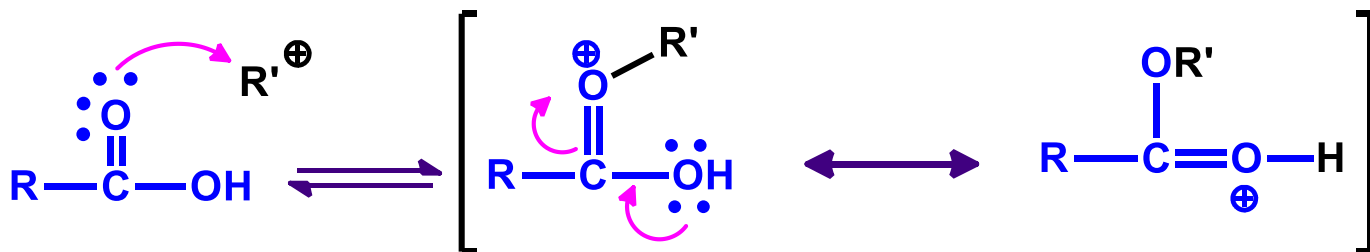
烷氧键断裂机理 (一般适用于叔醇的酯化)

1 正离子的形成

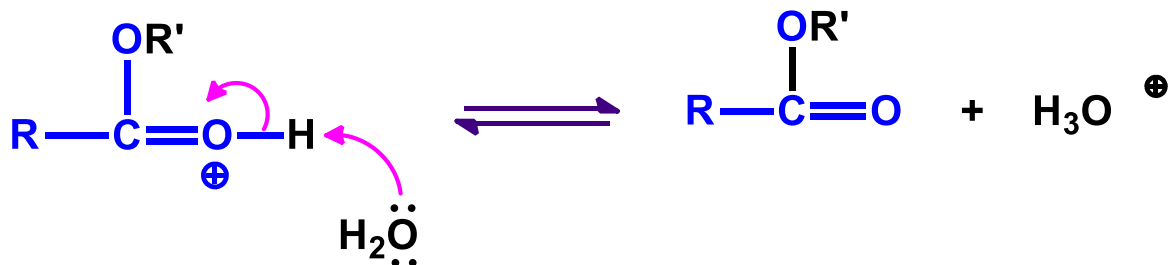


2 酸碱复合物的形成

与酰氧键断裂第1步相似。



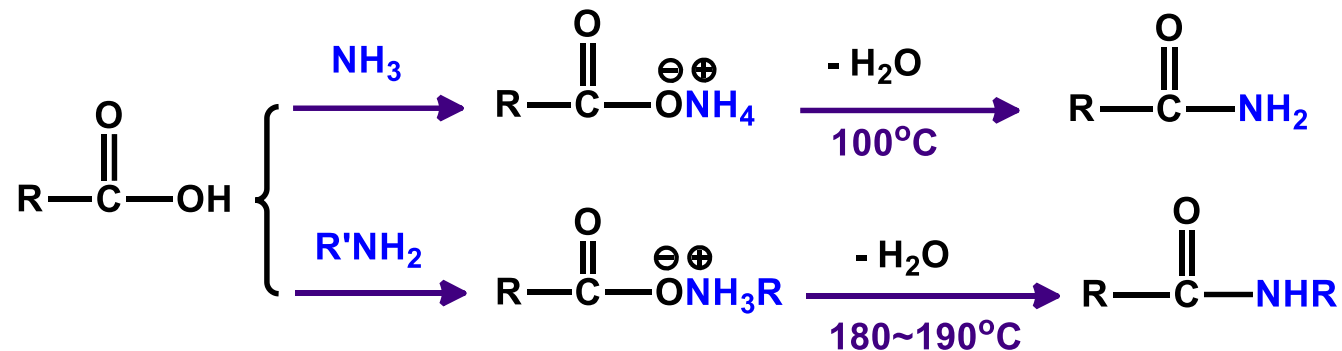
3 脱质子



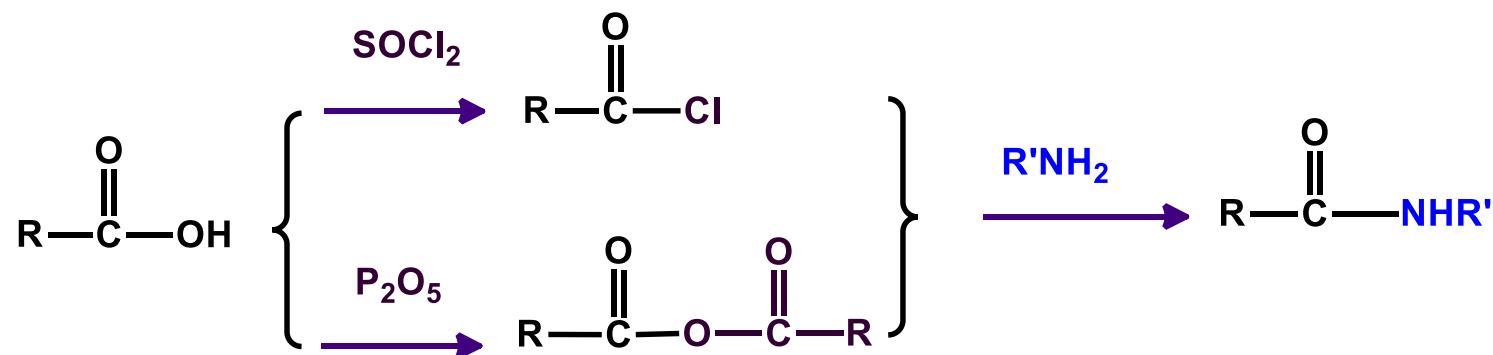


9.4 羧酸的化学性质

羧酸衍生物的生成-酰胺



➤ 合成上一般通过羧酸衍生物制备酰胺

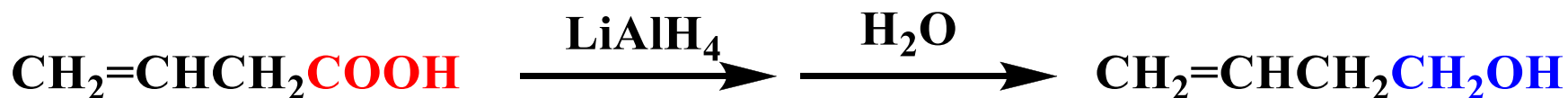
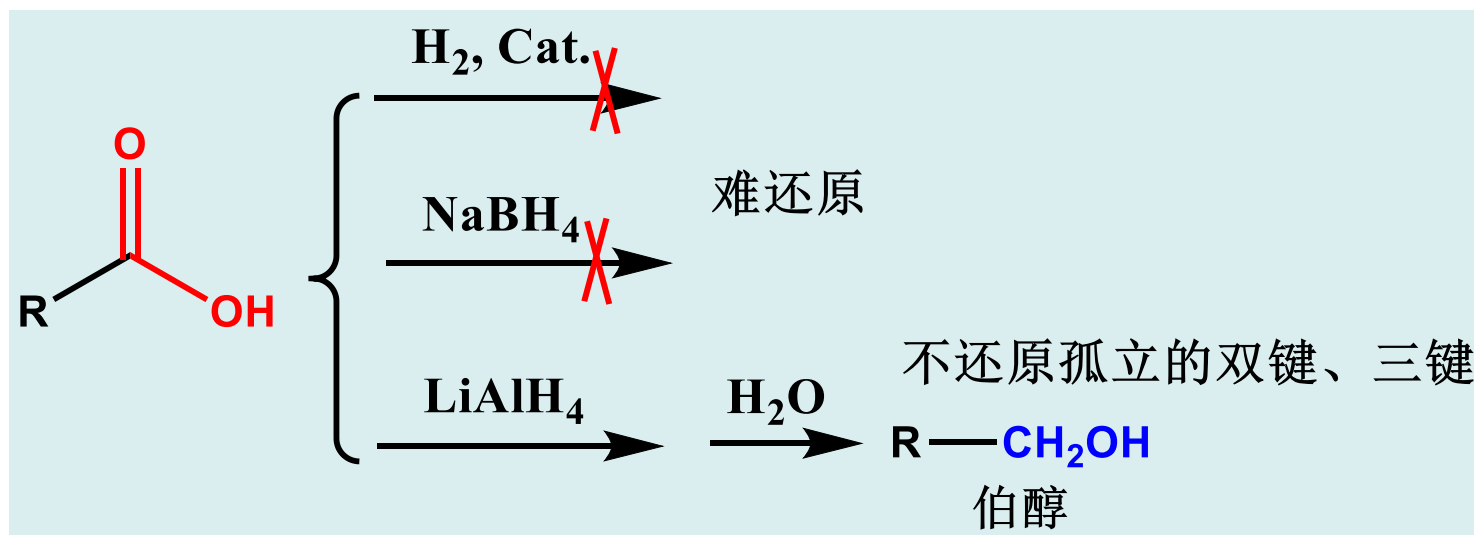


9.4 羧酸的化学性质



9.4.3 还原反应

- ✓ 羧基较难被还原，不能被催化氢化和 NaBH_4 还原
- ✓ 强还原剂 (LiAlH_4) 可还原至伯醇；分子中的双键一般不受影响



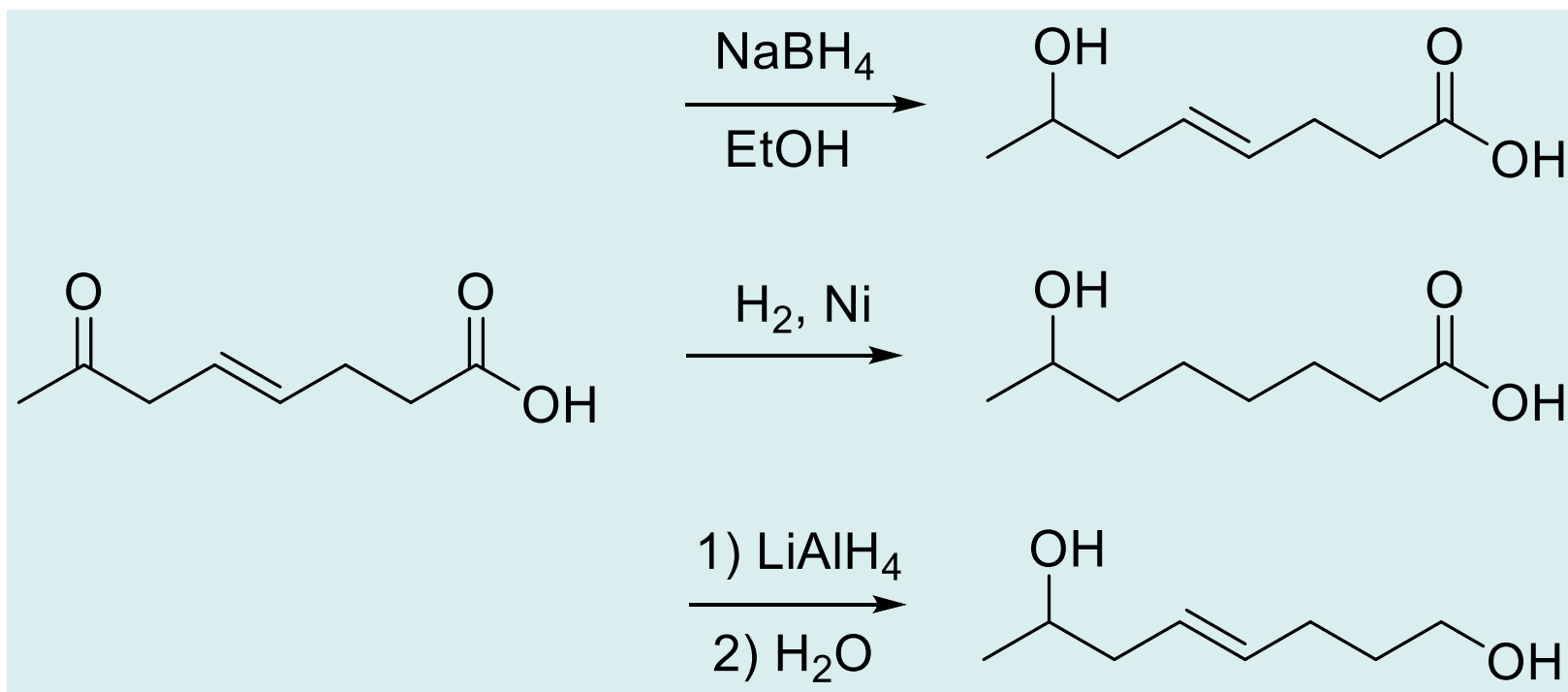
- ✓ 乙硼烷 (B_2H_6) 也可将羧酸还原成伯醇



9.4 羧酸的化学性质

官能团的选择性还原

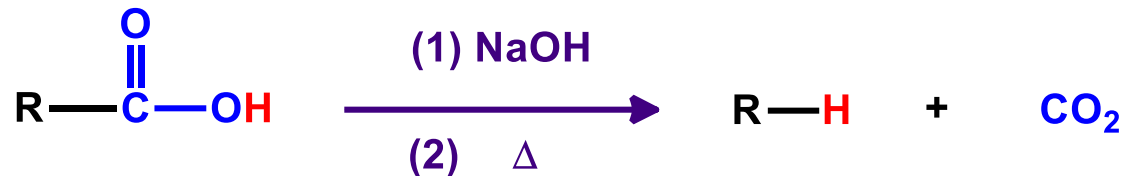
- ✓ NaBH_4 一般只还原醛、酮，不能还原羧酸
- ✓ 催化氢化可还原醛酮、烯烃、炔烃等
- ✓ LiAlH_4 既可还原醛酮，也可还原羧酸及其衍生物——强还原剂



9.4 羧酸的化学性质

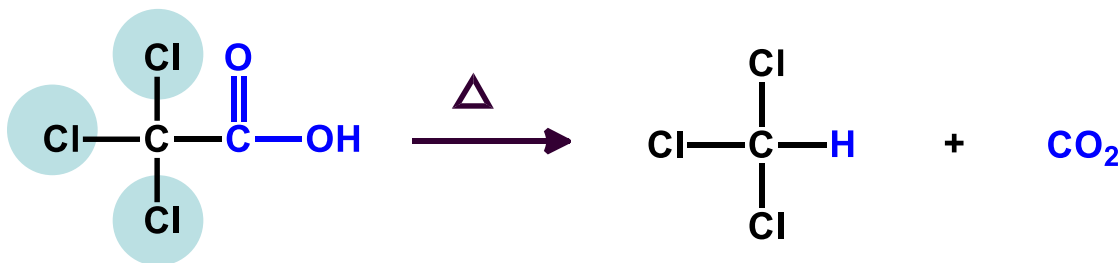


9.4.4 脱羧反应

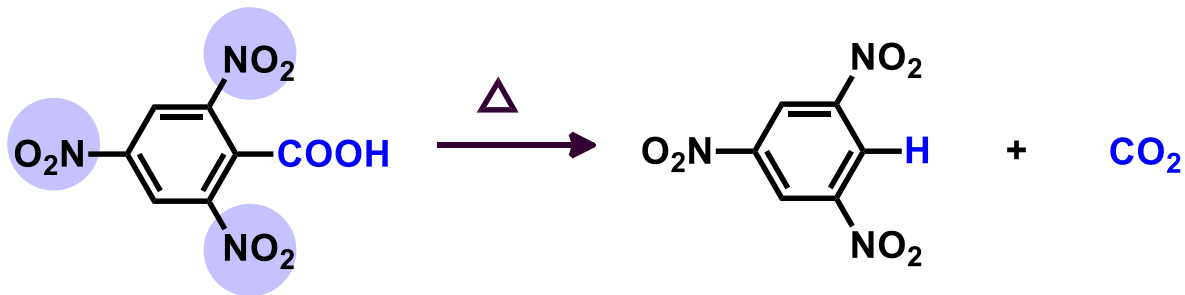


R有吸电基团时容易发生: $-\text{N}^+ \begin{smallmatrix} \text{O} \\ \parallel \\ \text{O}^- \end{smallmatrix}$ $-\text{C}\equiv\text{N}$ $\begin{smallmatrix} \text{O} \\ \parallel \\ \text{R} \end{smallmatrix}$ $-\text{Cl}$

例 1:



例 2:

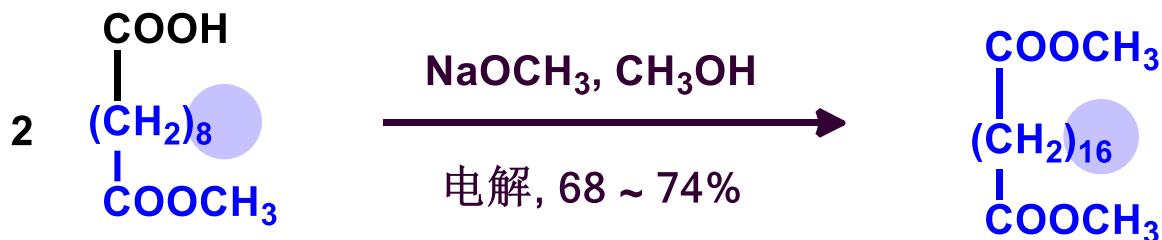




9.4 羧酸的化学性质

9.4.4 脱羧反应

Kolbe合成法：电解羧酸盐溶液,在阳极发生烷基的偶联生成烃。
反应可在水或甲醇中进行。



9.4 羧酸的化学性质



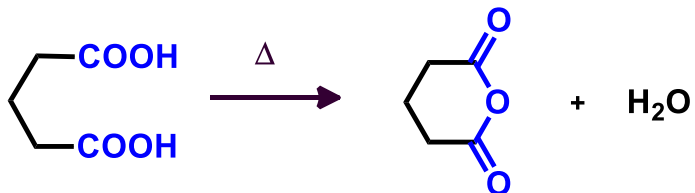
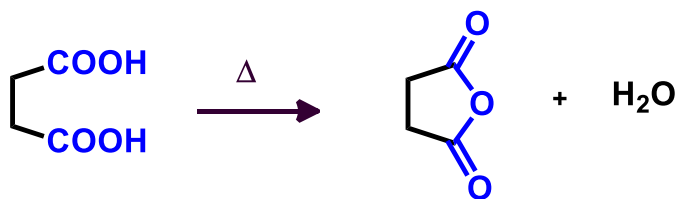
9.4.5 二元酸受热反应

1

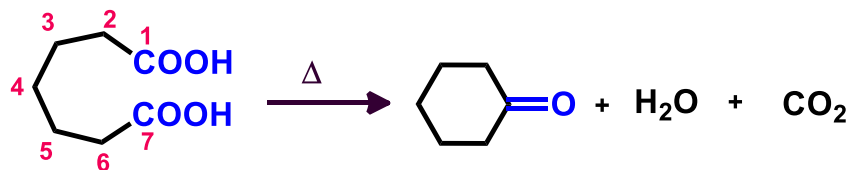
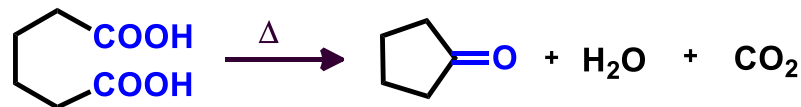


Blanc 规律： 容易形成五元环或六元环化合物。

2



3

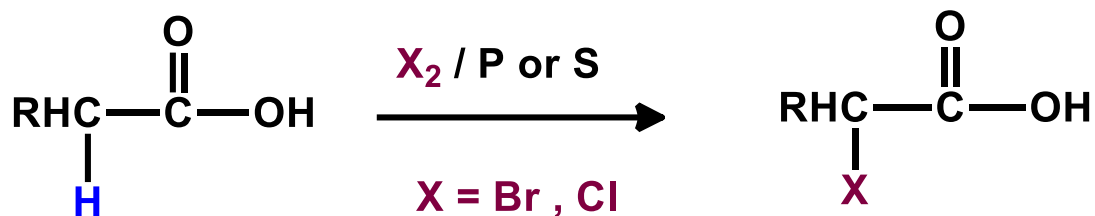




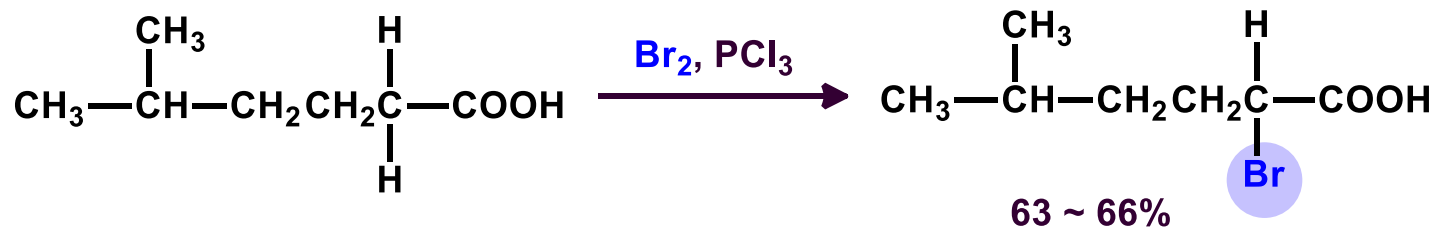
9.4 羧酸的化学性质

9.4.6 α -氢原子的反应

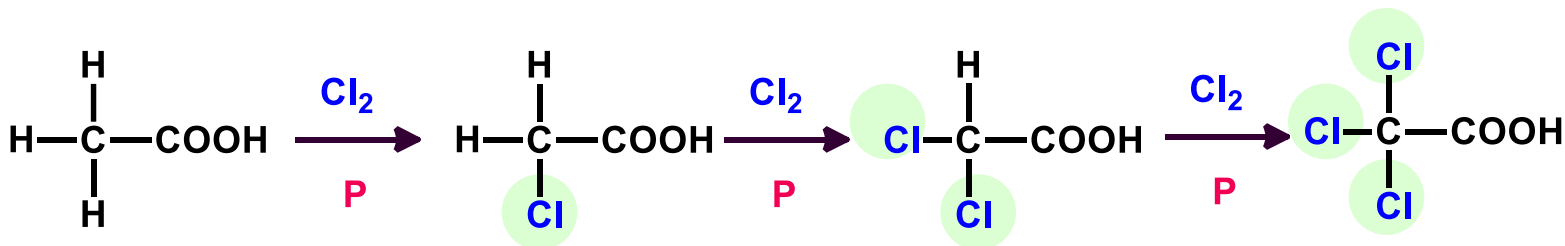
Hell-Volhard-Zelinsky 反应



例 1:



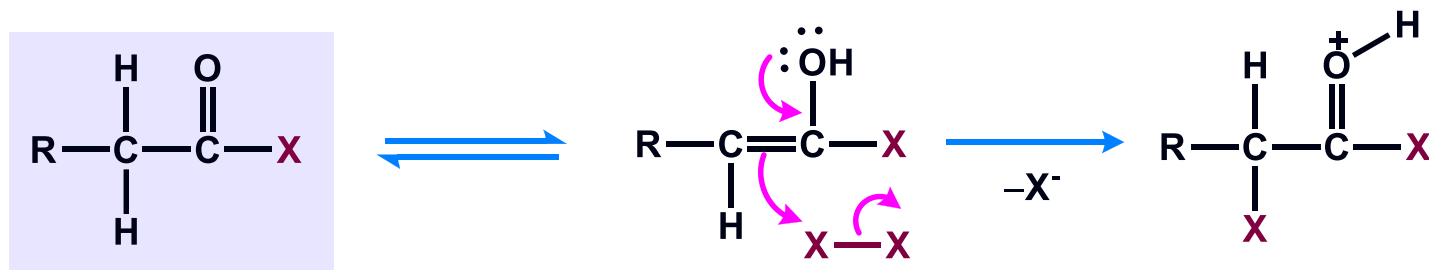
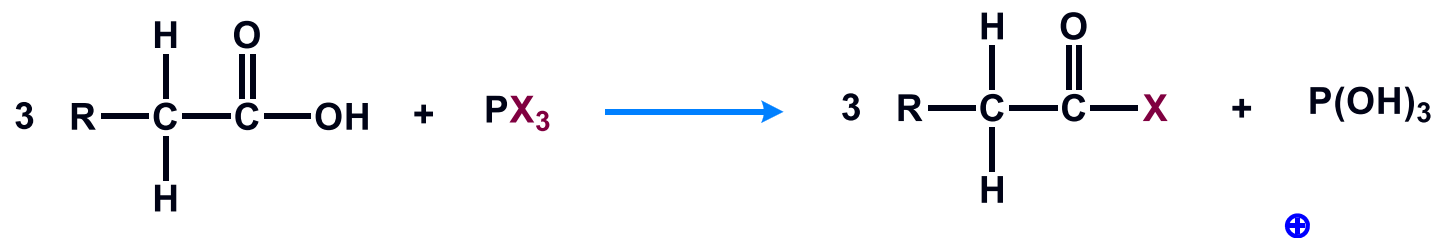
例 2:





9.4 羧酸的化学性质

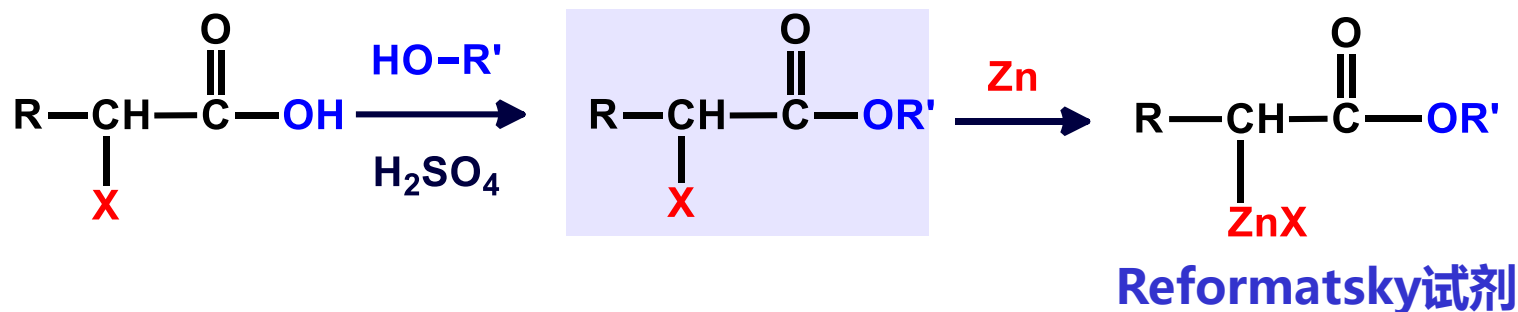
Hell-Volhard-Zelinsky 反应机理



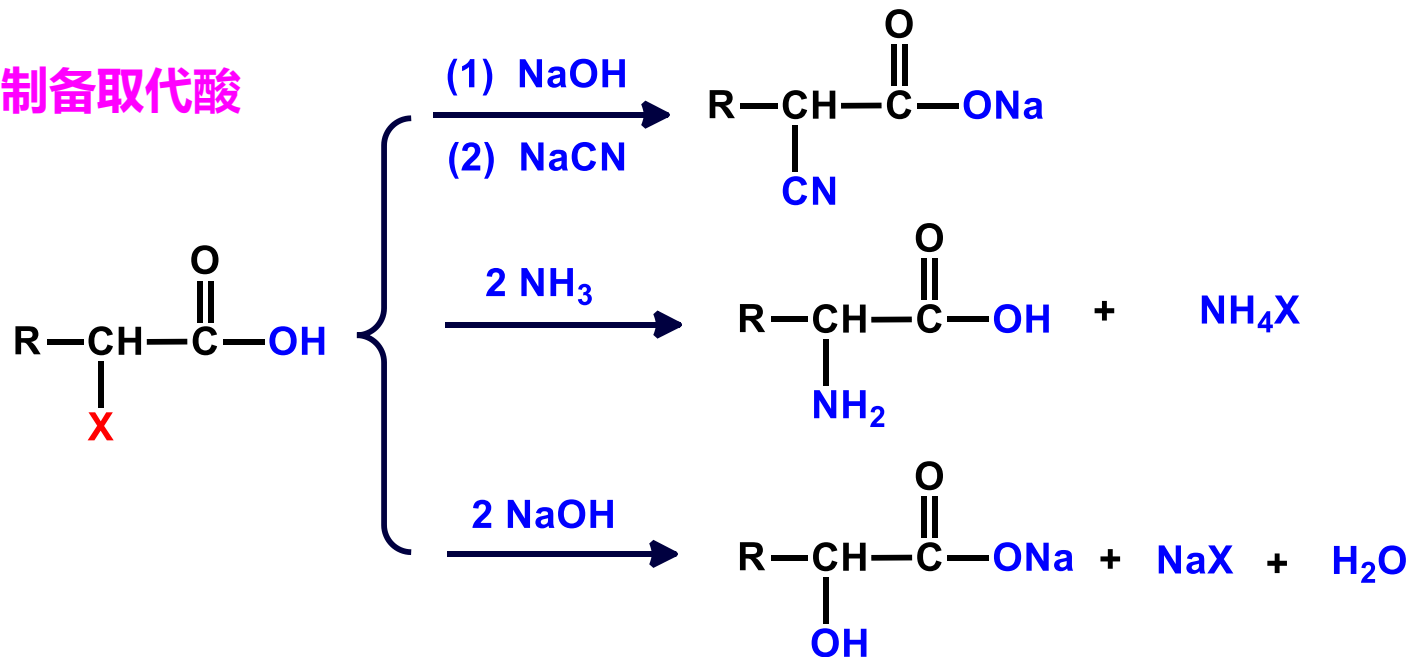


9.4 羧酸的化学性质

(i) 制备 α -卤代酯 (制备Reformatsky反应的原料)



(ii) 制备取代酸



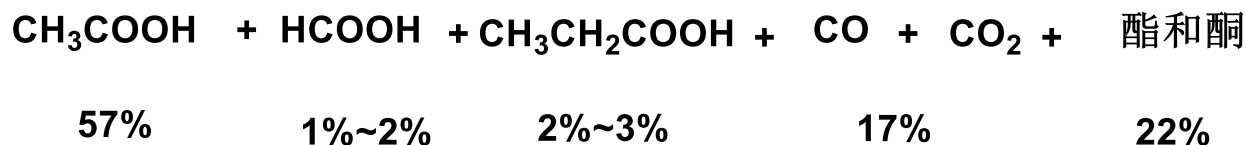
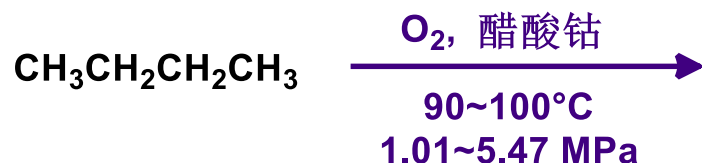


9.5 羧酸的制备

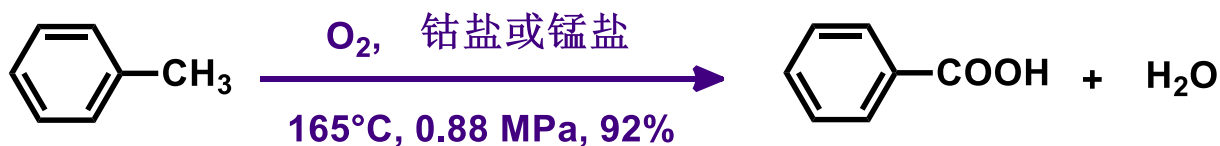
9.5.1 羧酸的工业合成——(1) 烃的氧化

了解

烷烃



芳烃



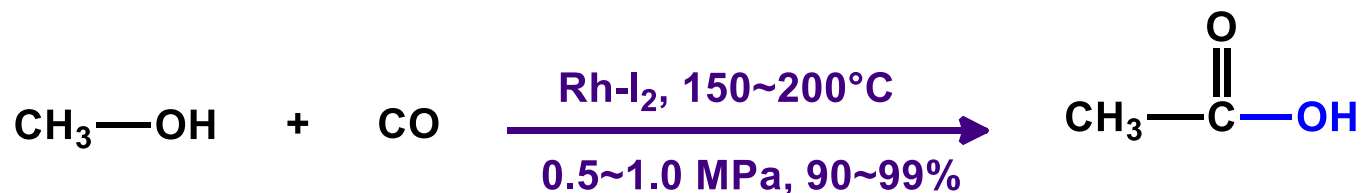
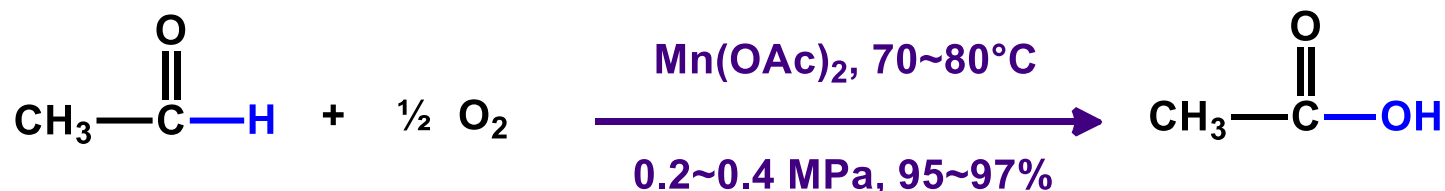
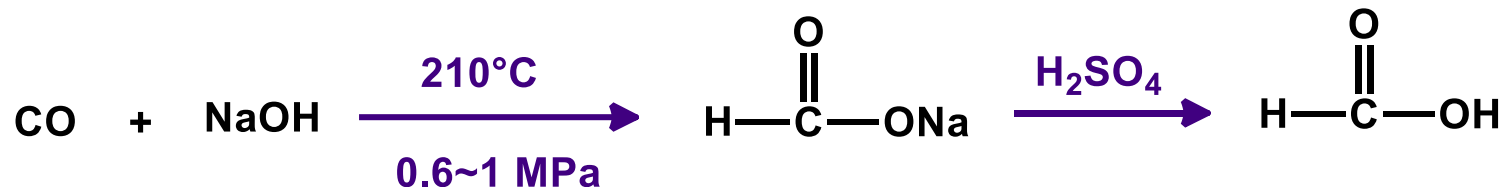


9.5 羧酸的制备

9.5.1 羧酸的工业合成

了解

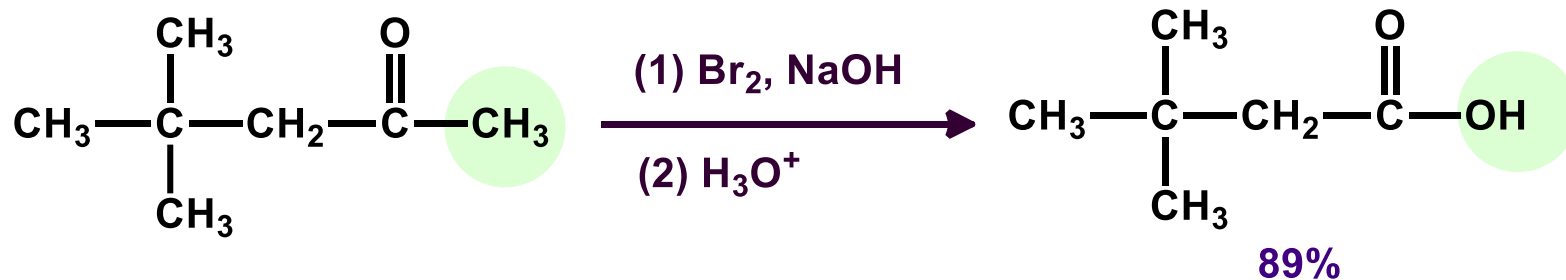
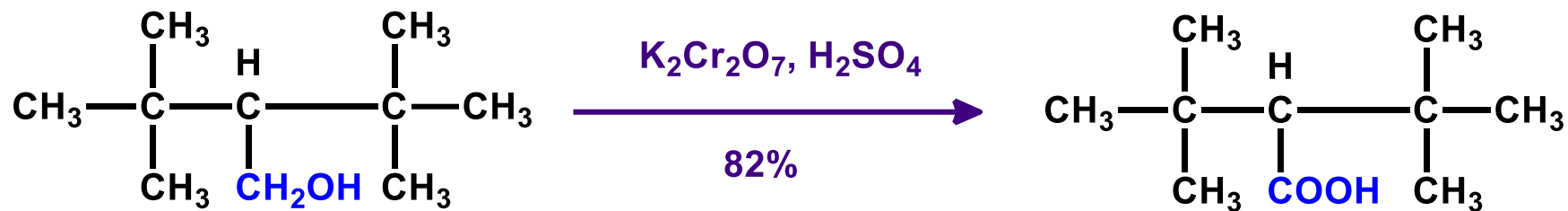
由一氧化碳、甲醇或醛制备



9.5 羧酸的制备



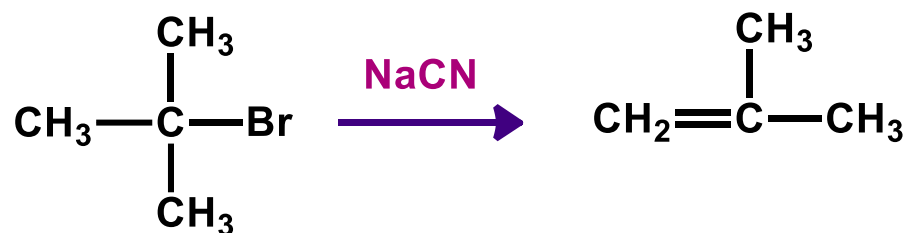
9.5.2 伯醇和醛的氧化



9.5 羧酸的制备



9.5.3 腈的水解

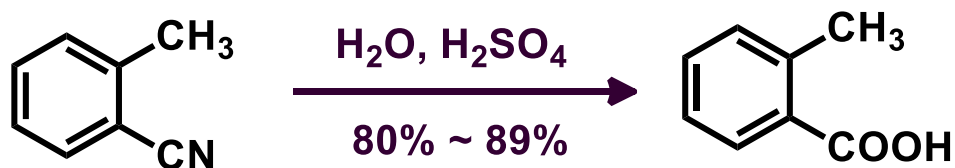
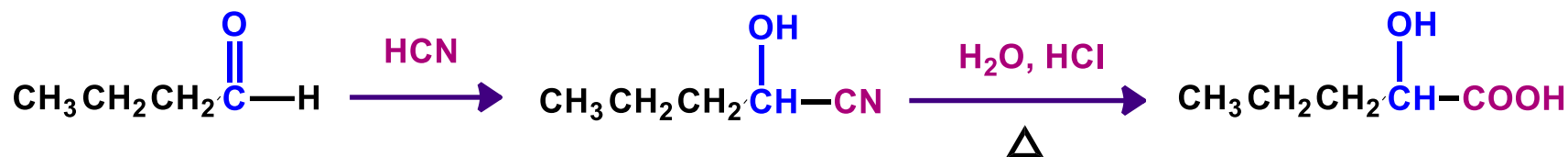


叔卤代烃一般不用于此反应。

9.5 羧酸的制备



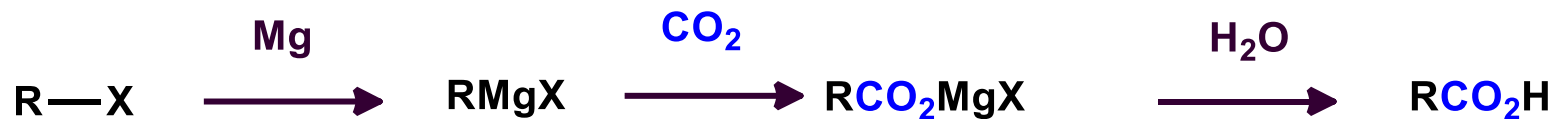
9.5.3 腈的水解



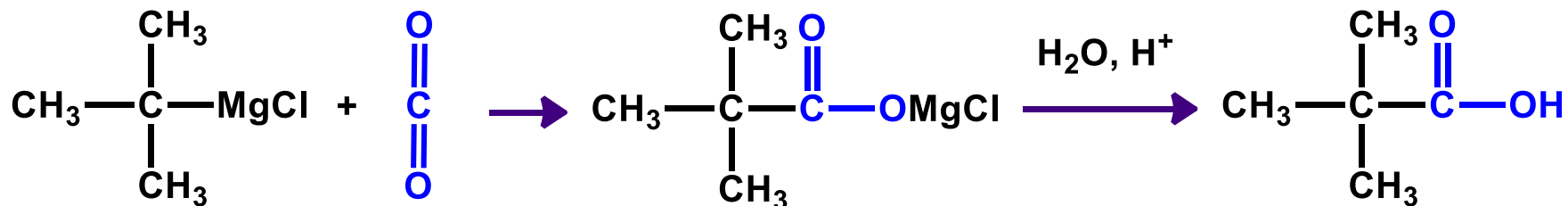
9.5 羧酸的制备



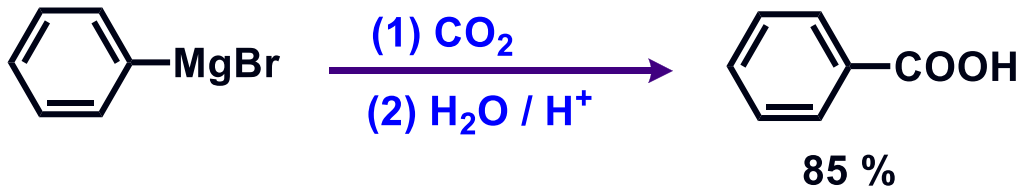
9.5.4 Grignard试剂与二氧化碳作用



例:



79% ~ 80%

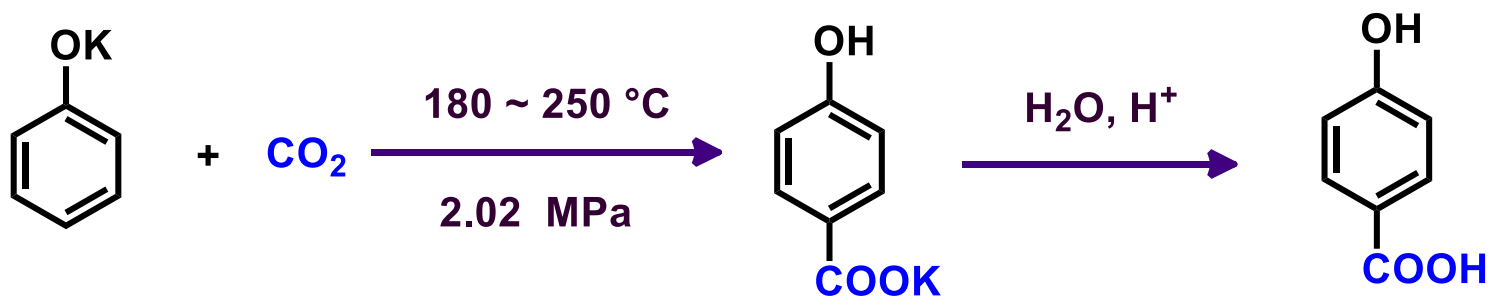
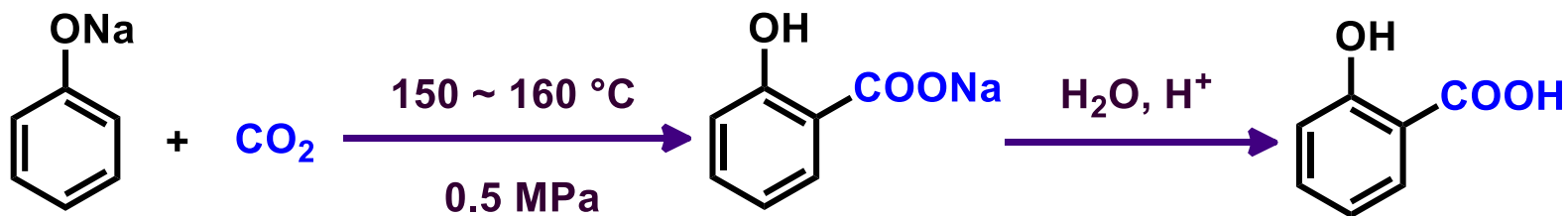


9.5 羧酸的制备



9.5.5 酚酸的合成

Kolbe-Schmitt反应



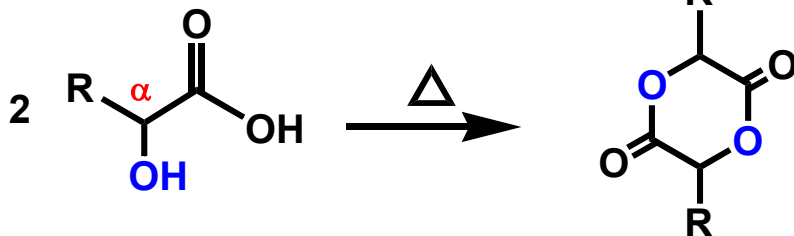
9.6 羧酸



9.6.1 羧酸的脱水

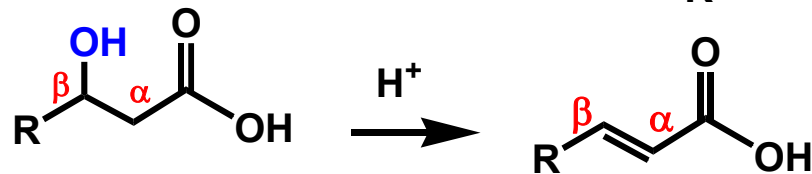
了解

α -羟基酸



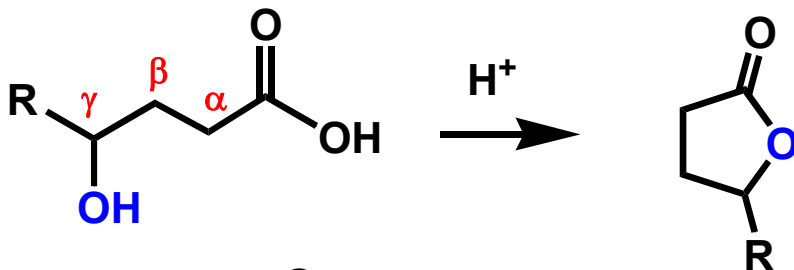
交酯

β -羟基酸



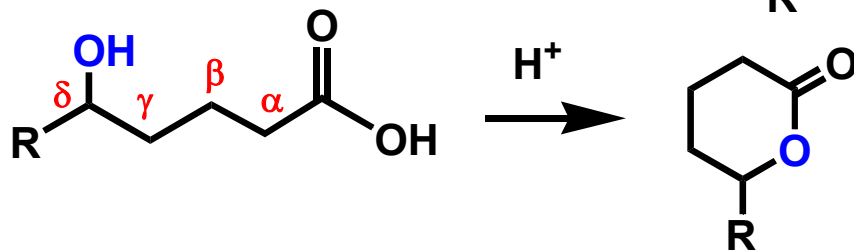
α, β -不饱和酸

γ -羟基酸



γ -内酯

δ -羟基酸



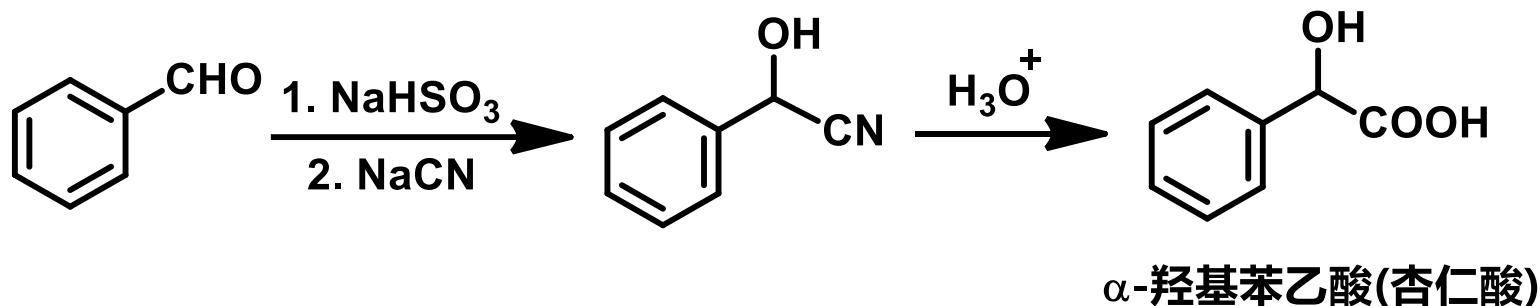
δ -内酯

9.6 羧酸



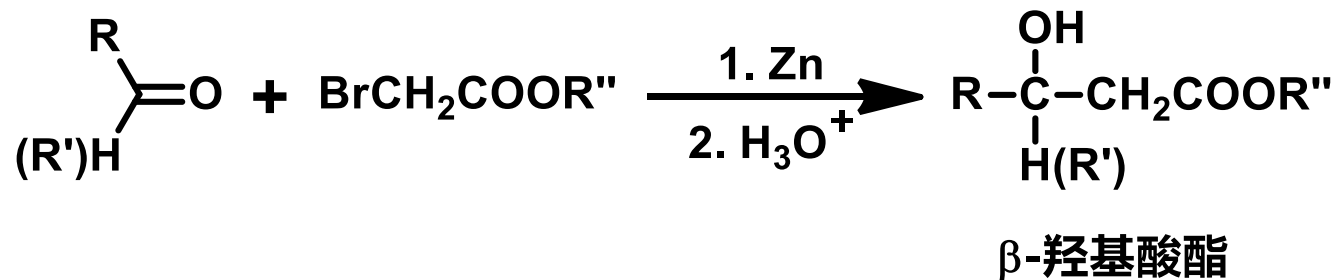
9.6.3 羧酸的制备

□ 水解法： α -卤代酸或 α -羟基腈



□ 瑞佛马斯基(Reformatsky)反应

➤ 醛或酮与 α -溴代酸酯和锌在惰性溶剂中相互作用





课后习题

- **作业：**
 - P412 练习 12.7 & 12.8
 - P418 练习 12.12
 - P419 习题（四） & （六）
-
- **作业要求：**
 - 使用统一的作业本，**先抄一遍题目再作答。**
 - 下次课前完成，随机抽查。
 - **下次作业提交时统一批改给分后计入平时分。**
 - 课代表收齐后交至助教



练习题

